

ECSE-4040

Digital Electronics

Lecture 17

Bipolar Digital Circuits

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References: Appendix B: HJS

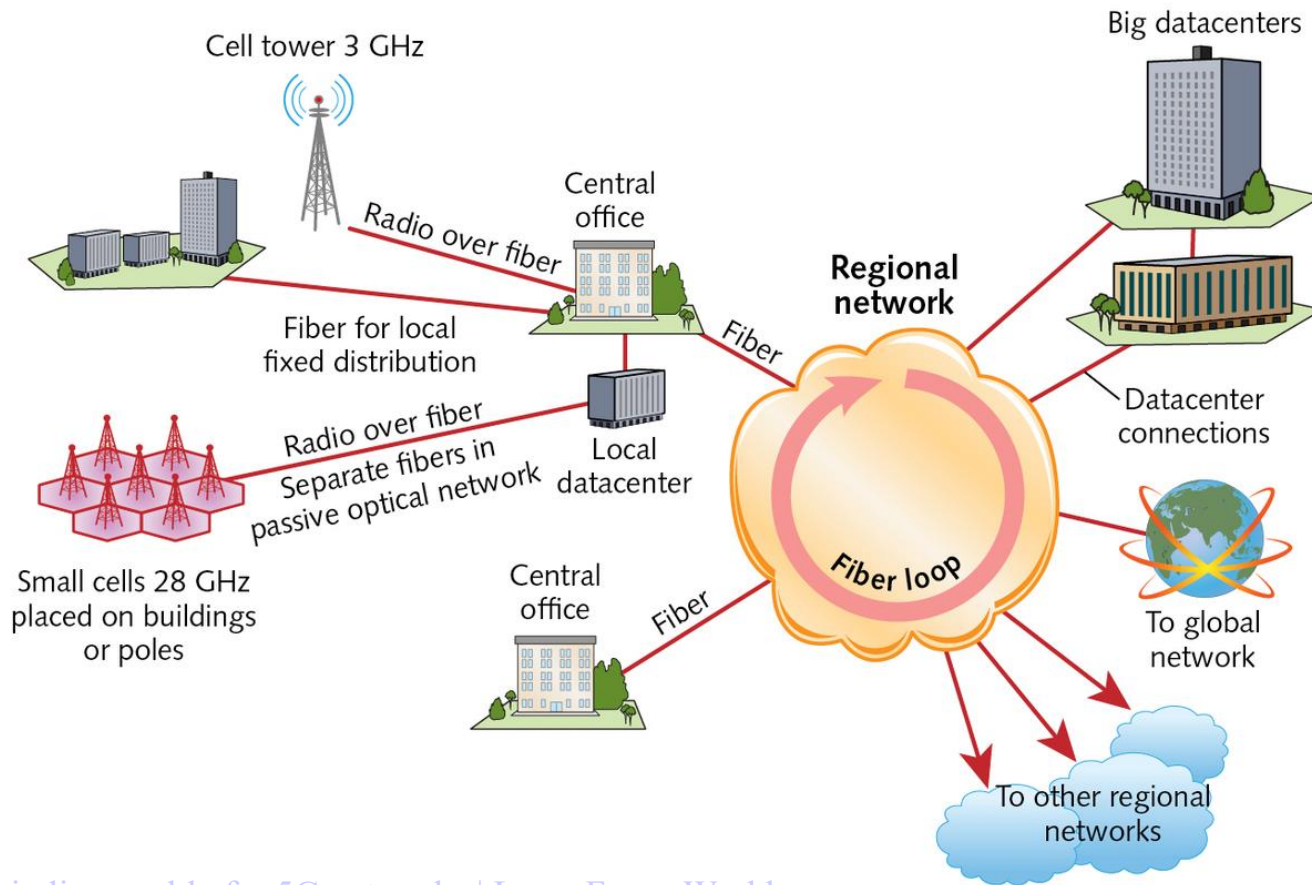
Chapters 5 and 17, Neamen

<https://lms.rpi.edu>

Lecture Outline

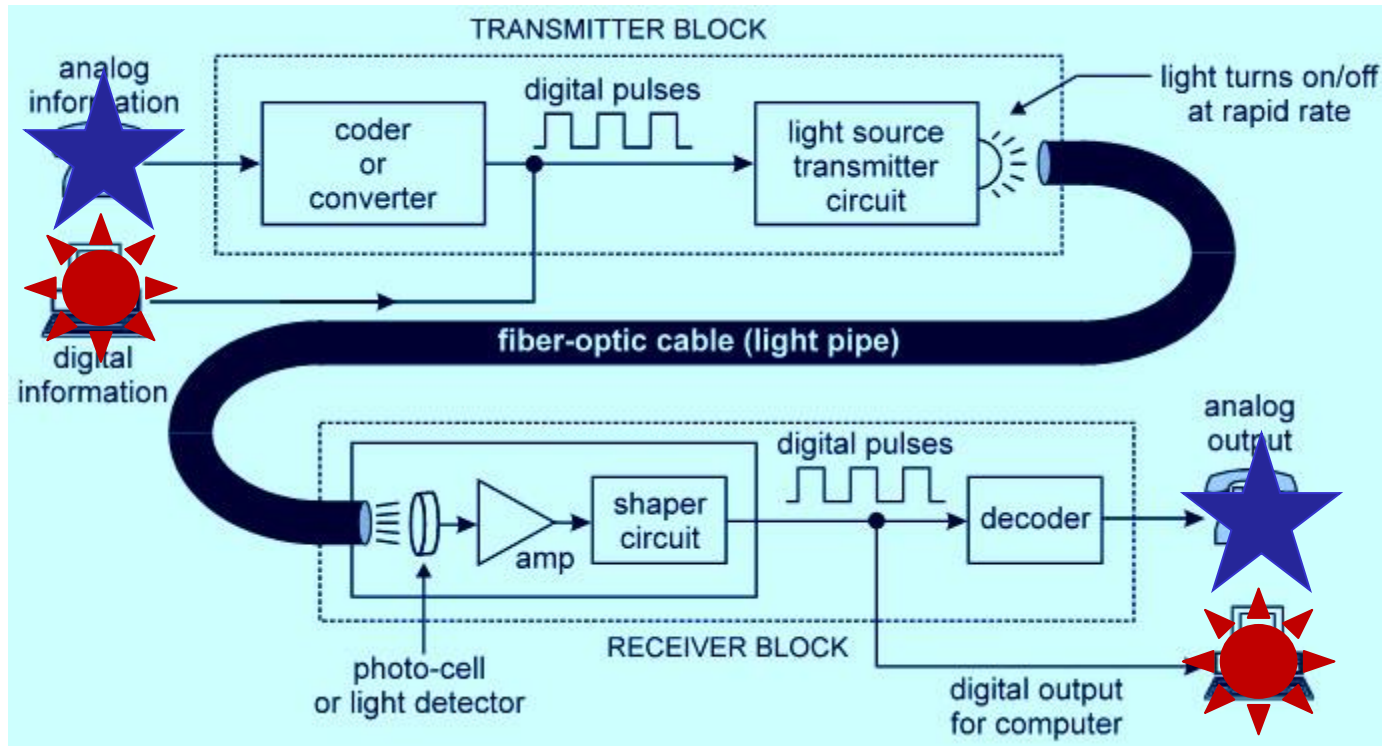
- Why BJT in digital electronics?
- Review BJT Basics
- BJT SPICE Representation
- VTC of a BJT Inverter
- DC Analysis
- Example of digital circuits

Digital ICs for Communications applications



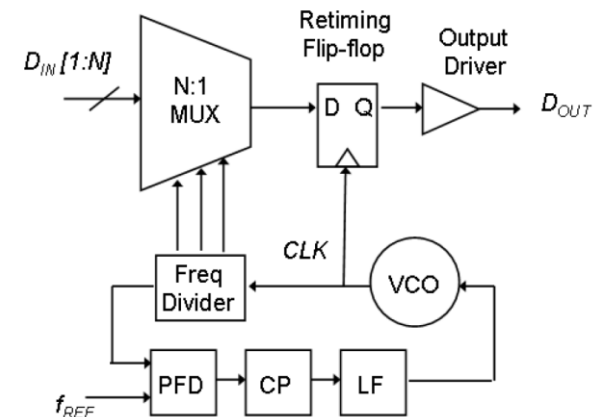
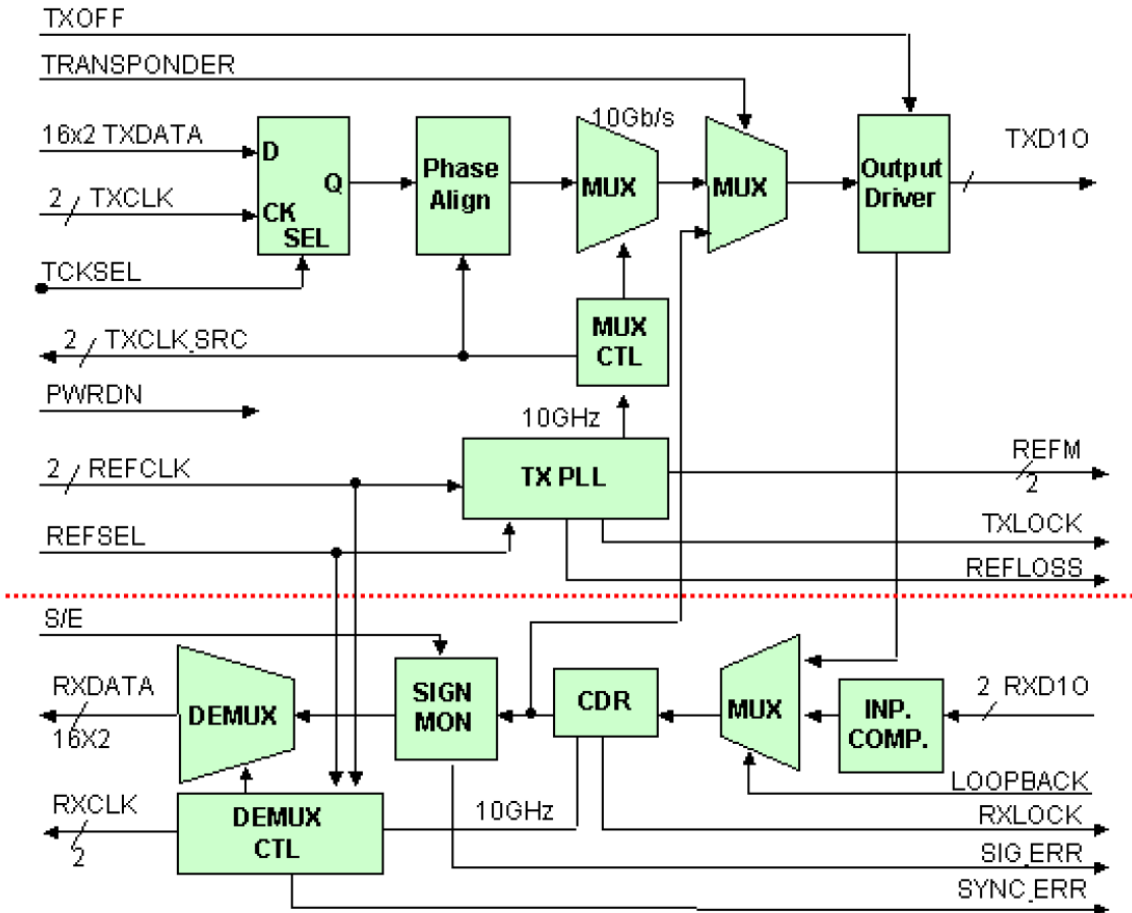
[Optics will be indispensable for 5G networks | Laser Focus World](#)

Inside a Transceiver for Fiber-Optic Communication System



Examples of High Speed Digital ICs

Fiber-Optic SERDES



Serializer

High Speed Logic Families

Emitter-Coupled Logic (ECL)

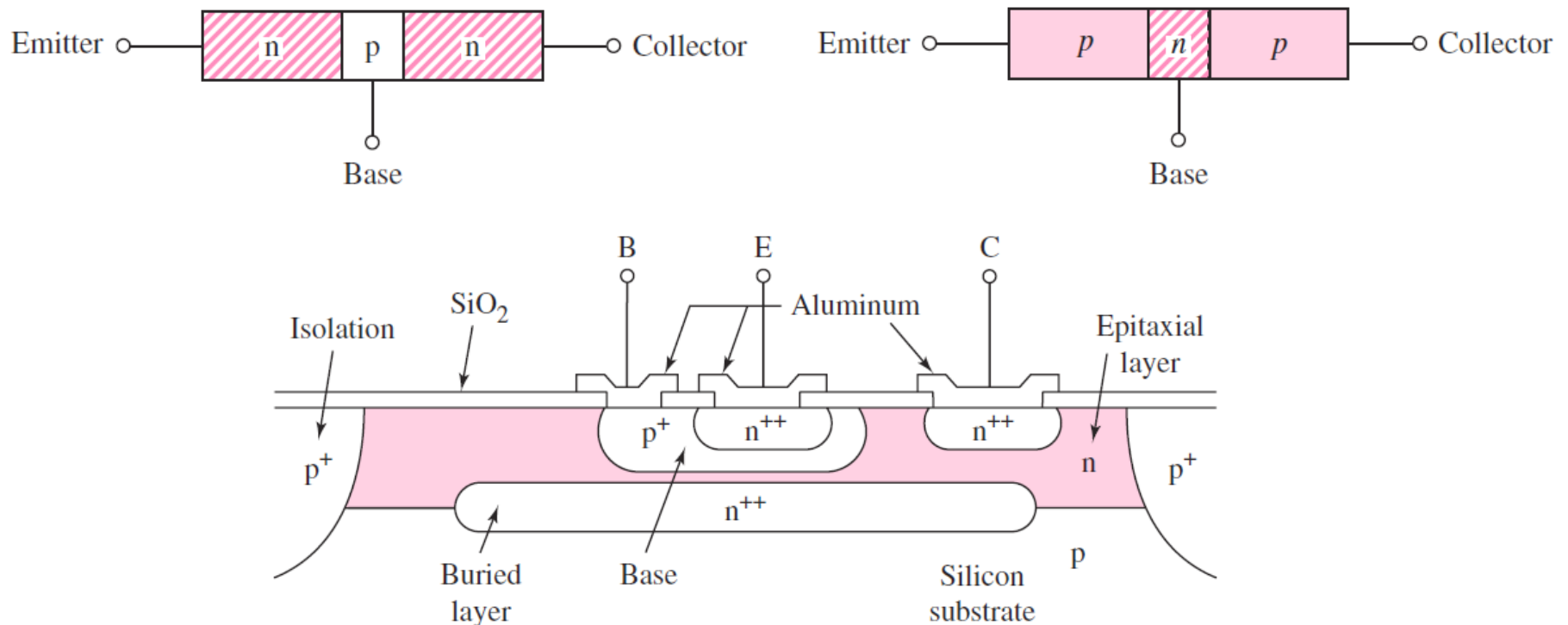
- Bipolar
- III-V FET
- BiCMOS

Current-Mode Logic (CML)

- ☐ Bipolar
- ☐ MOS
- ☐ BiCMOS

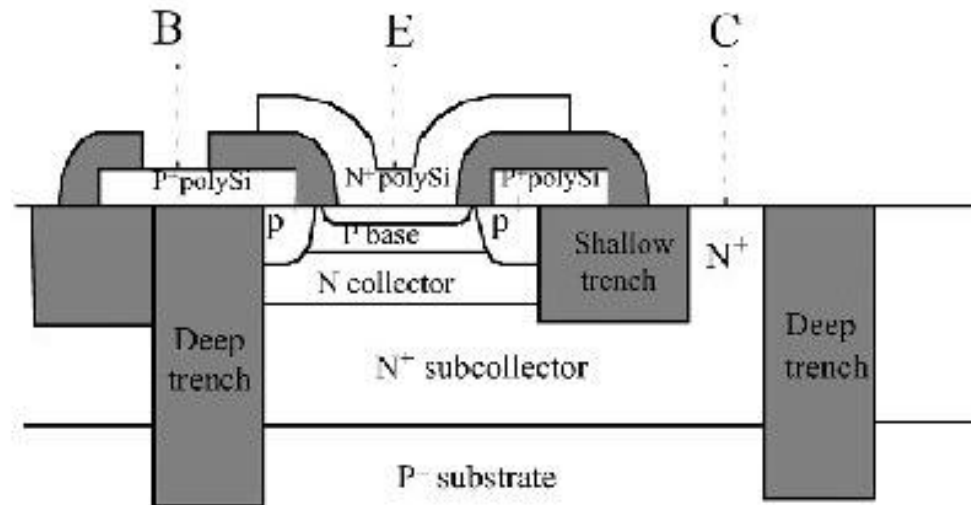
****Some high-speed logic families use differential topologies**

Review: Bipolar Junction Transistor



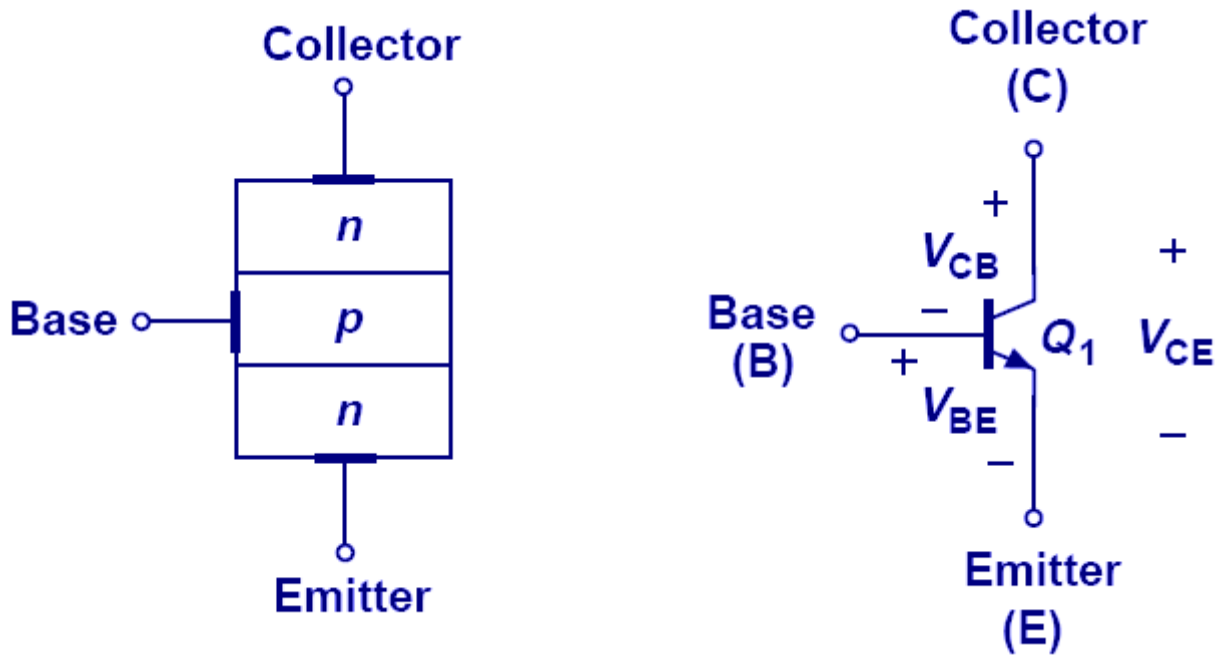
Cross section of a conventional integrated circuit npn bipolar transistor

Modern BJT Structure



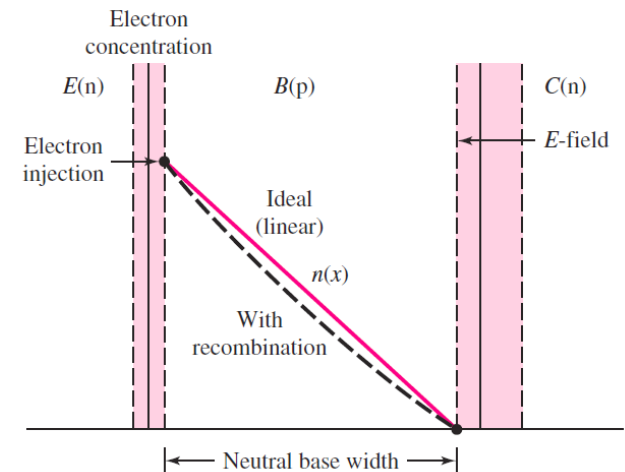
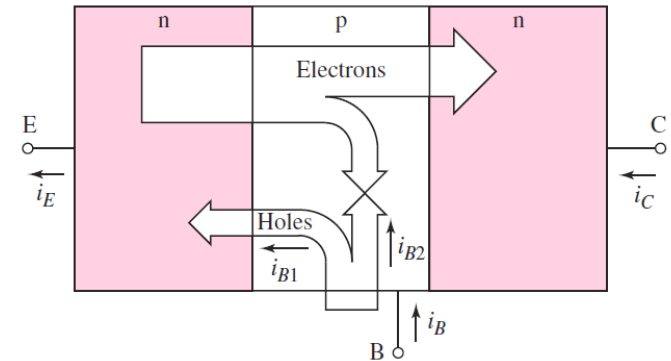
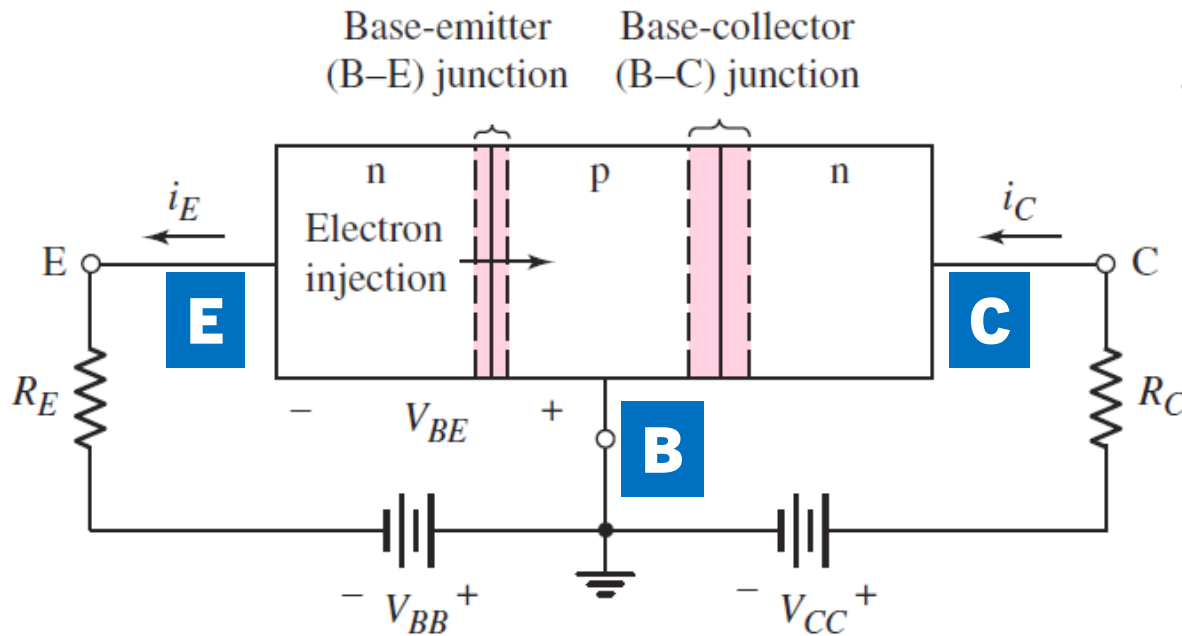
- Narrow base
- n+ poly-Si emitter
- Self-aligned p+ poly-Si base contacts
- Lightly-doped collector
- Heavily-doped epitaxial subcollector
- Shallow trenches and deep trenches filled with SiO_2 for electrical **isolation**

NPN-BJT



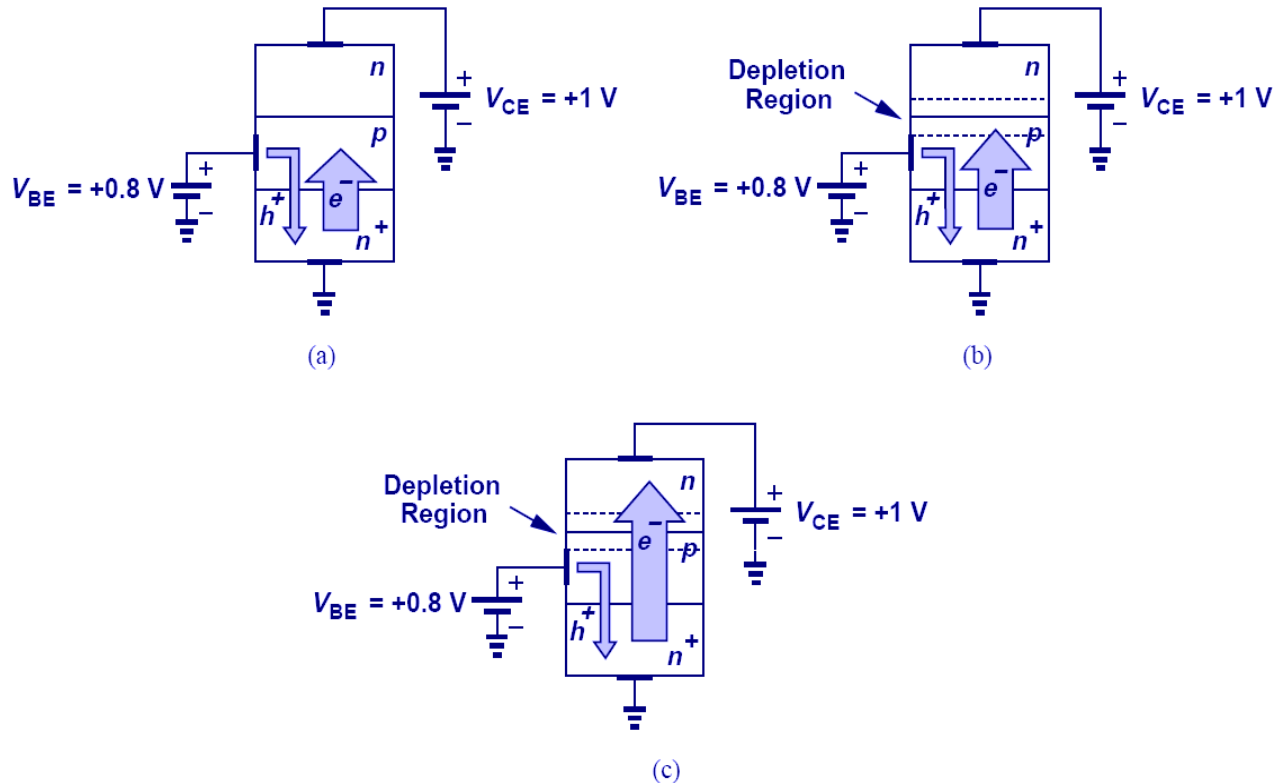
Two p-n junctions (B-E & B-C) $\rightarrow$ 4 regions of operation

Regions of Operation for NPN Transistor: 1. Forward Active



- Forward active region: $V_{BE} > 0$, $V_{BC} < 0$.

Accurate Bipolar Representation



- Collector also carries current due to carrier injection from base.

Collector Current

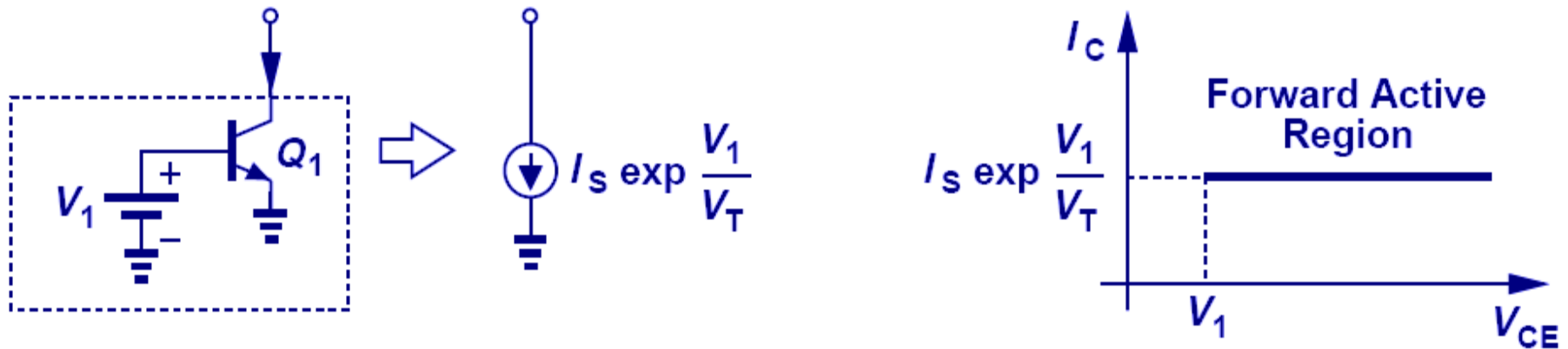
$$I_C = \frac{A_E q D_n n_i^2}{N_E W_B} \left(\exp \frac{V_{BE}}{V_T} - 1 \right)$$

$$I_C \approx I_S \exp \frac{V_{BE}}{V_T}$$

$$I_S = \frac{A_E q D_n n_i^2}{N_E W_B}$$

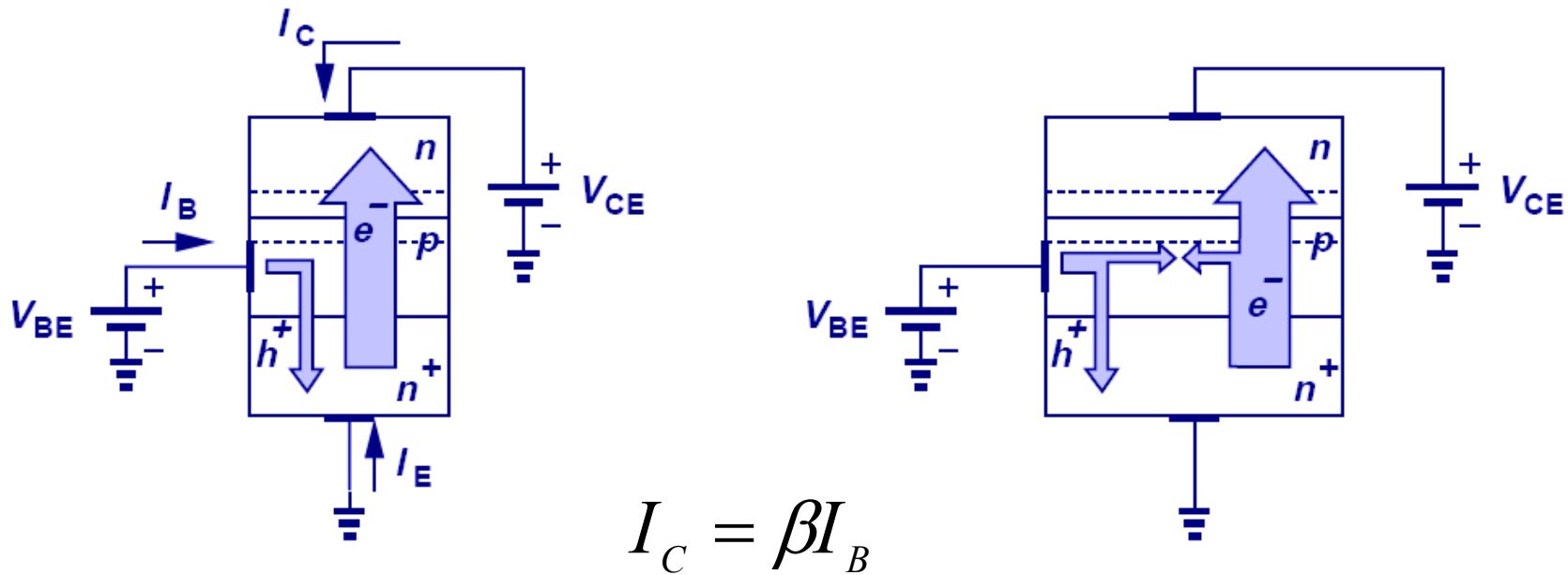
- Applying the law of diffusion, we can determine the charge flow across the base region into the collector.

¹³ Forward Active Region: Constant Current Source



- Ideally, the collector current does not depend on the collector to emitter voltage. This property allows the transistor to behave as a constant current source when its base-emitter voltage is fixed.

Forward Active Region: What happened to the Base Current?



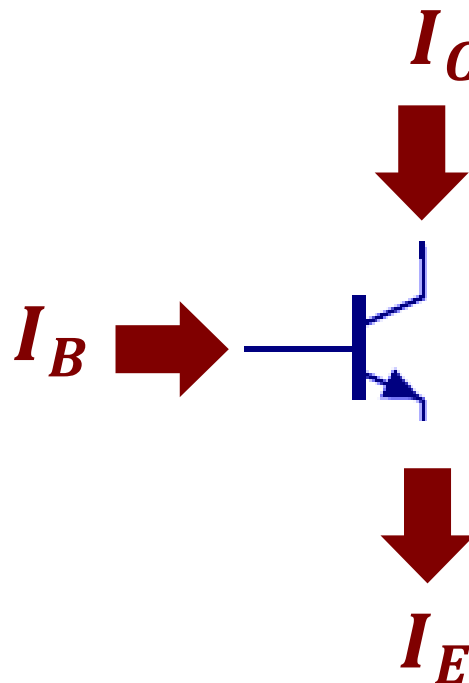
- Base current consists of two components:
 - 1) Reverse injection of holes into the emitter
 - 2) recombination of holes with electrons coming from the emitter.

Forward Active Region: Emitter Current

$$I_E = I_C + I_B$$

$$I_E = I_C \left(1 + \frac{1}{\beta} \right)$$

$$\beta = \frac{I_C}{I_B}$$



- Applying Kirchoff's current law to the transistor, we can easily find the emitter current.

Summary of Currents

$$I_C \approx I_S \exp \frac{V_{BE}}{V_T}$$

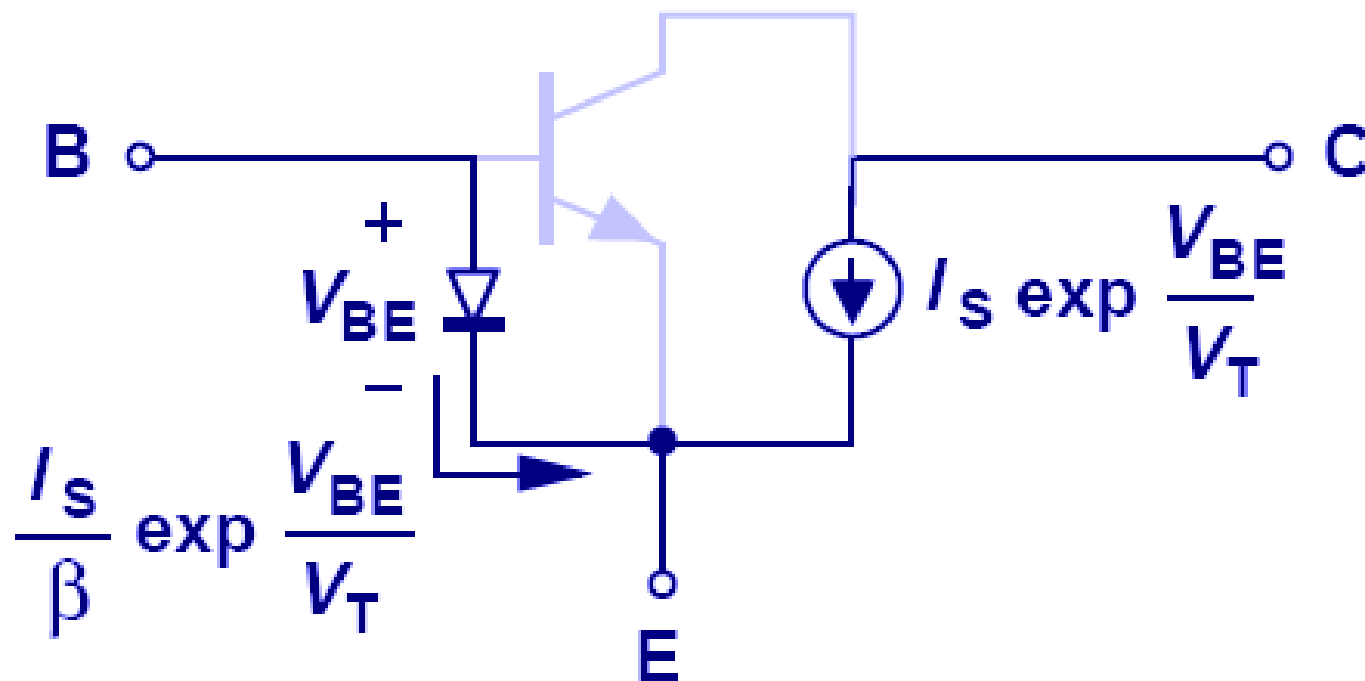
$$I_B \approx \frac{1}{\beta} I_S \exp \frac{V_{BE}}{V_T}$$

$$I_E \approx \frac{\beta + 1}{\beta} I_S \exp \frac{V_{BE}}{V_T}$$

$$\frac{\beta}{\beta + 1} = \alpha$$

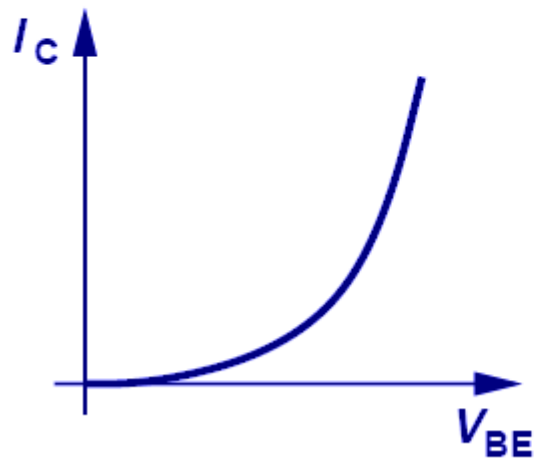
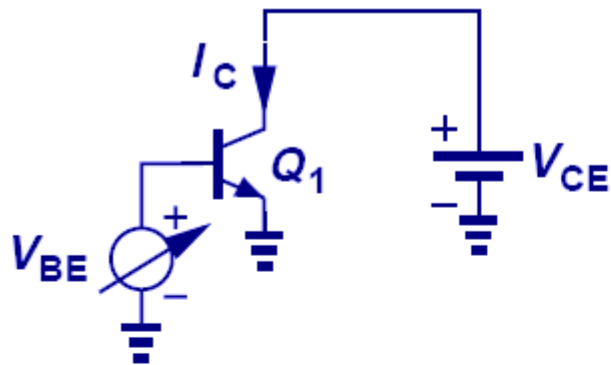
$$V_T = \frac{KT}{q}$$

Bipolar Transistor Large Signal Model—Forward Active Mode

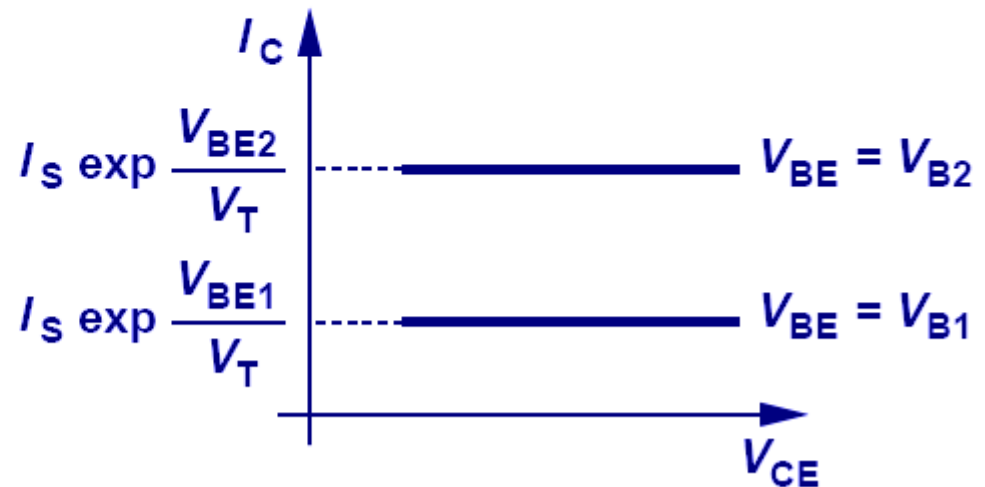
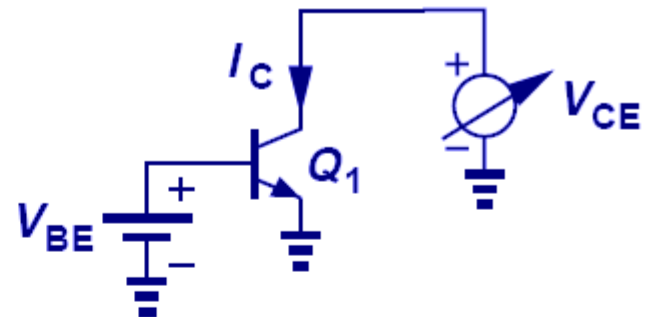


A diode is placed between base and emitter and a voltage controlled current source is placed between the collector and emitter.

Characteristics of Bipolar Transistor : Forward Active Mode

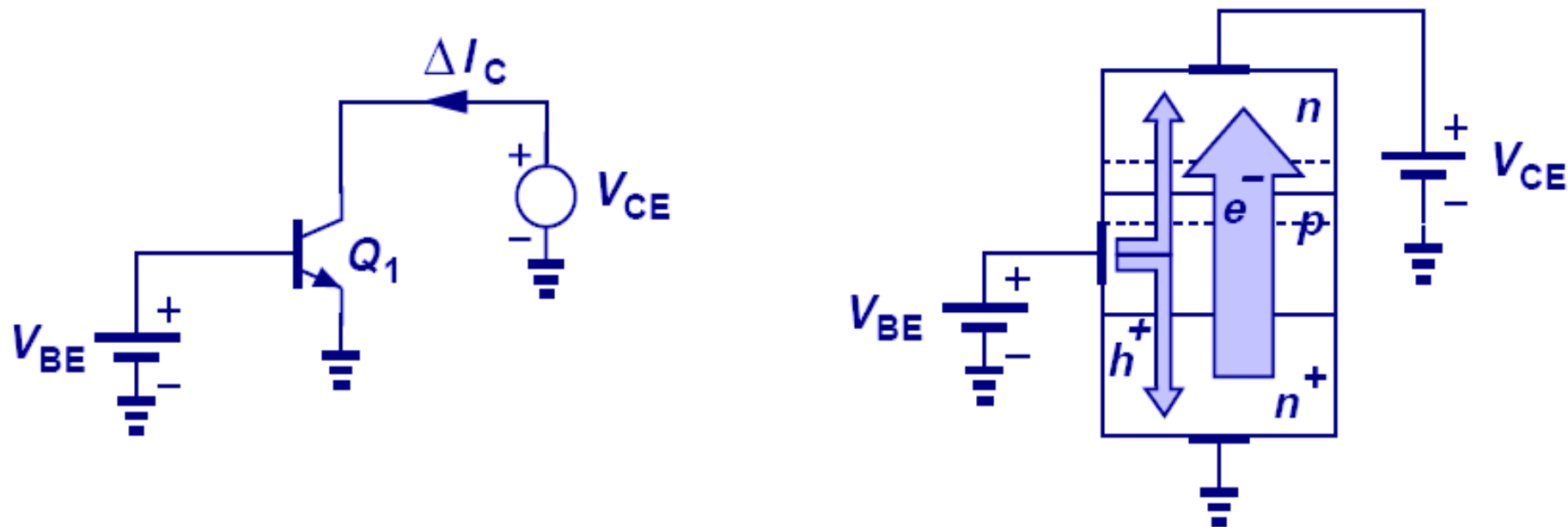


(a)



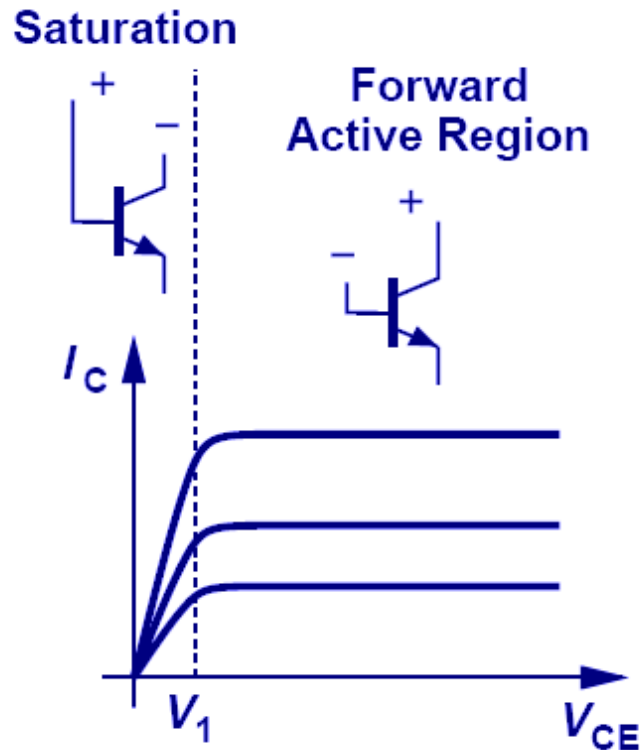
(b)

Bipolar Transistor in Saturation



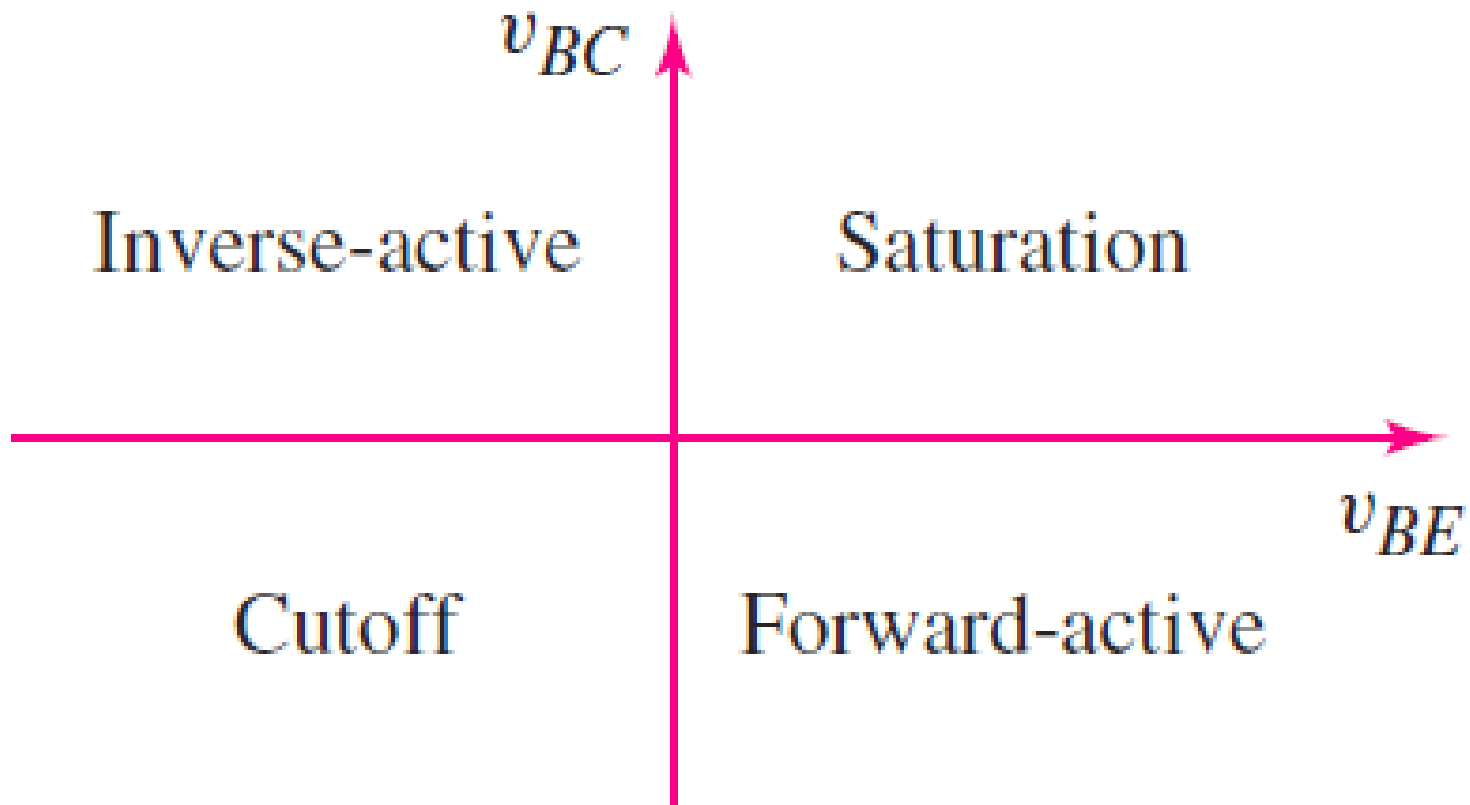
- When collector voltage drops below base voltage and forward biases the collector-base junction, base current increases and decreases the current gain factor, β .

Overall I/V Characteristics

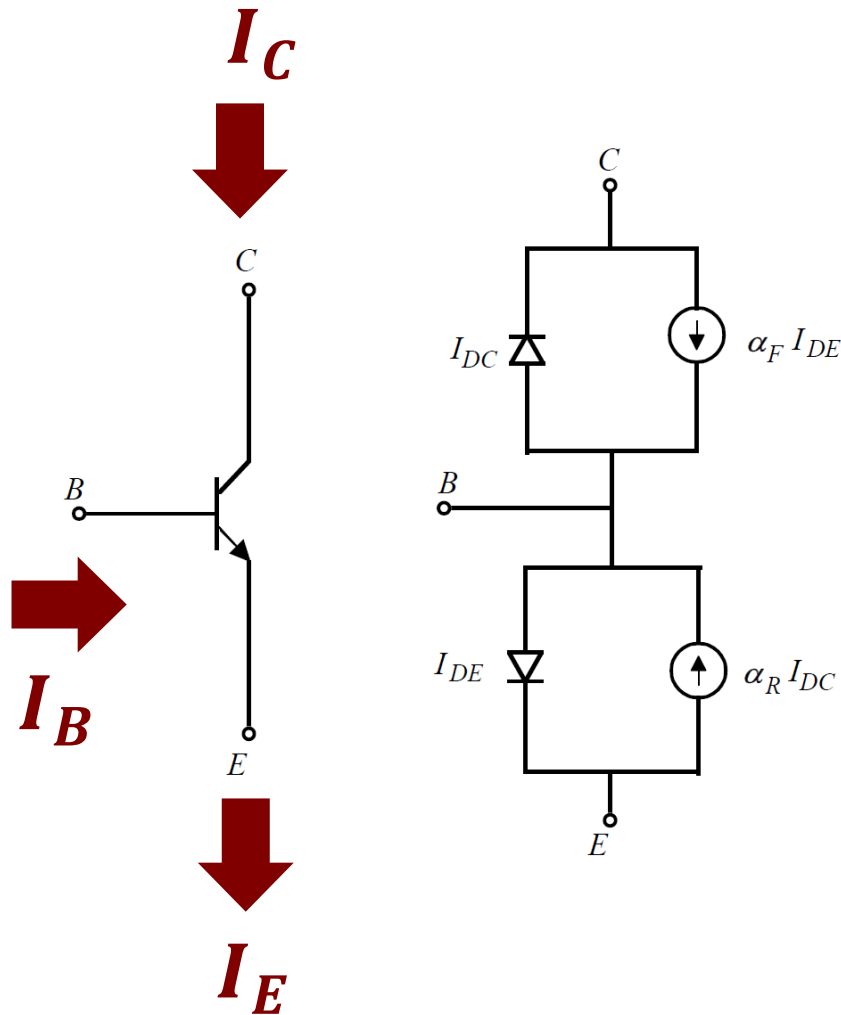


- The *speed* of the BJT also drops in saturation.

Summary of BJT Modes of Operation (NPN)



Ebers-Moll Model



Equations:

$$I_{DE} = I_{ES} [\exp(V_{BE}/V_T) - 1]$$

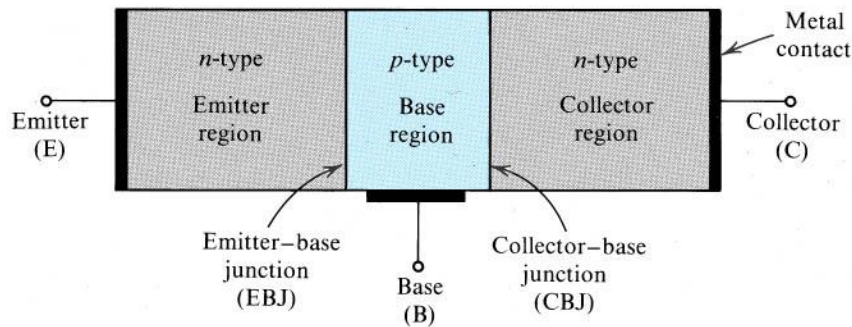
$$I_{DC} = I_{CS} [\exp(V_{BC}/V_T) - 1]$$

Typical values:

$$\alpha_F = .99 \quad I_{ES} = 10^{-15} \text{ A}$$

$$\alpha_R = .66 \quad I_{CS} = 10^{-15} \text{ A}$$

Summary of Modes of operation-Simplified Analysis



We have two junctions $\rightarrow$ 4 possible models of operation

1. Cut-off

B-E & B-C junctions are reverse biased.

$I_B = I_C = I_E = 0$ (ideal case, the actual value equals the reverse saturation current of the junction)

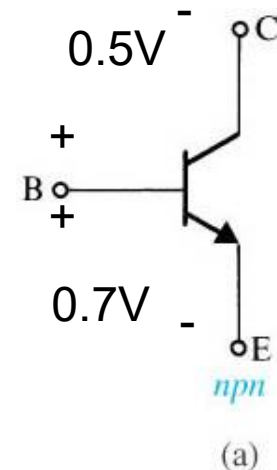
1. Saturation

If I_s is NOT given

B-E & B-C junctions are forward biased.

$$V_{BE} = 0.7, V_{BC} = 0.5, V_{CE} = 0.2V$$

$$I_E = I_C + I_B$$



Summary of Modes of operation

3. Forward Active

B-E junction is forward biased

B-C junction is reversed biased

Most electrons injected at the emitter go across the base to the collector

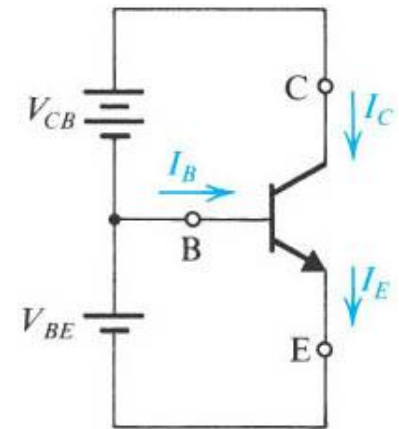
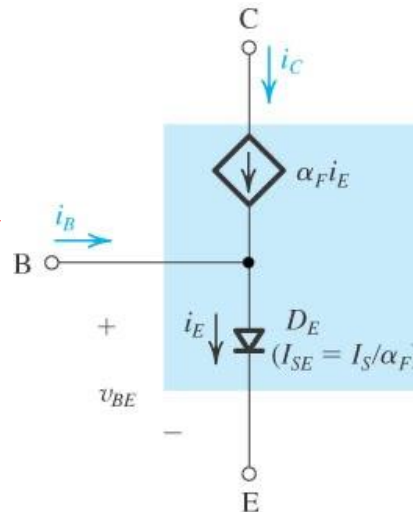
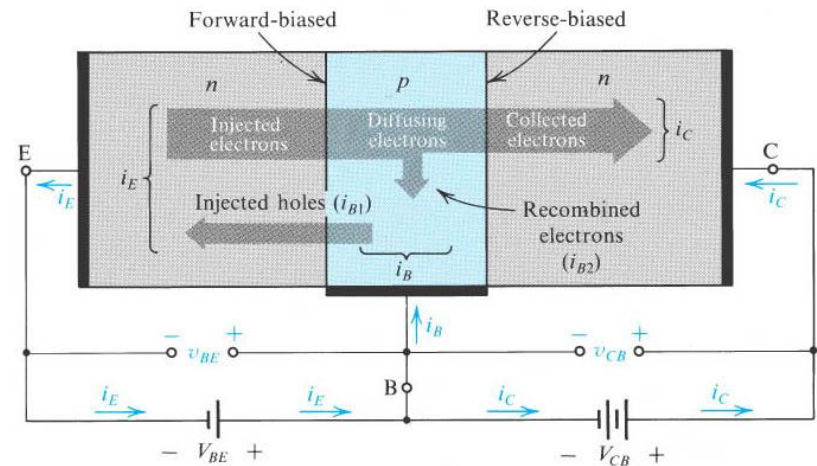
$I_C = \alpha_F I_E$ (α_F is the fraction of electrons going to the collector ~ 1 (0.96-0.99))

$$I_E = I_C + I_B$$

If I_S is not given

$$\frac{I_C}{I_B} = \beta_F, \frac{I_C}{I_E} = \alpha_F$$

$$\alpha_F = \frac{\beta_F}{\beta_F + 1}, \beta_F = \frac{\alpha_F}{1 - \alpha_F}$$



Summary of Modes of operation

4. Reverse Active

B-E junction is Reverse biased

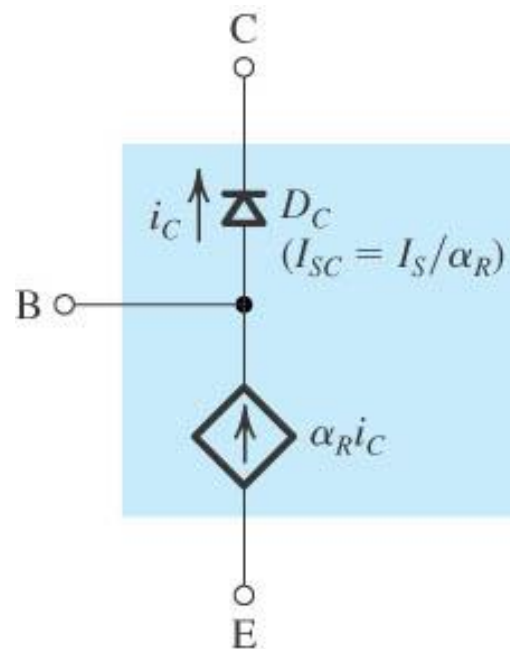
B-C junction is forward biased

$I_C = \alpha_R I_E$ (fraction of electrons from collector that reach the emitter $\ll 1$)

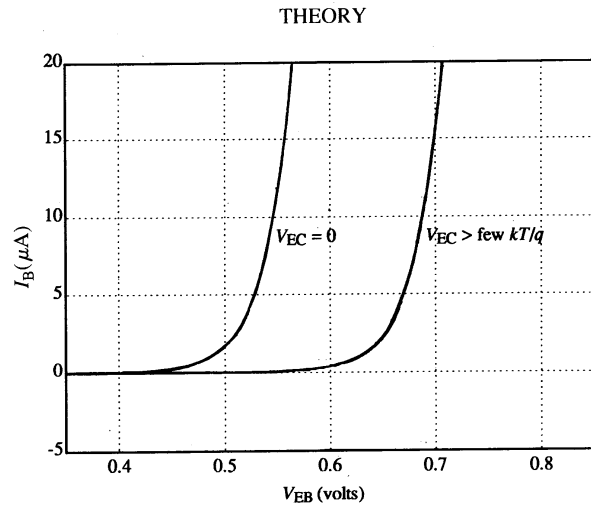
$$\frac{I_C}{I_B} = \beta_R, \frac{I_C}{I_E} = \alpha_R$$

$$\alpha_R = \frac{\beta_R}{\beta_R + 1}$$

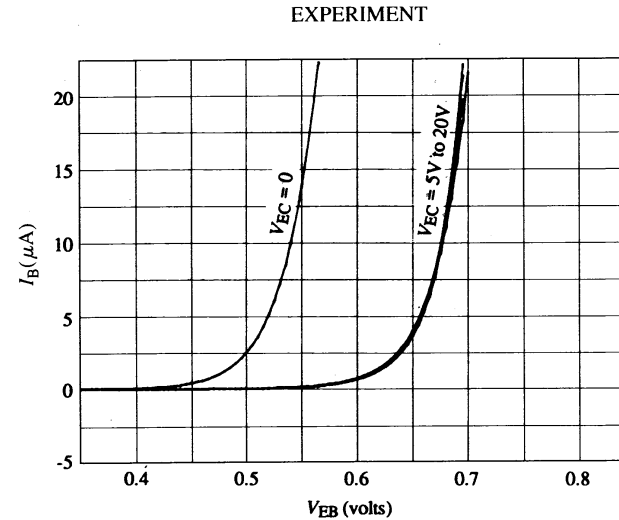
$$\beta_R \ll \ll \ll \beta_F$$



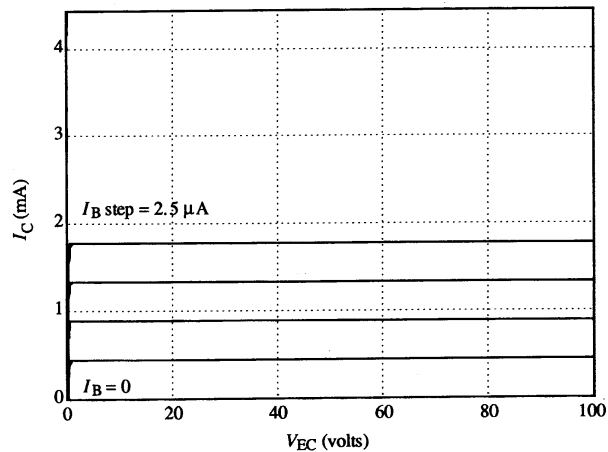
Comparing Theory and Experiments



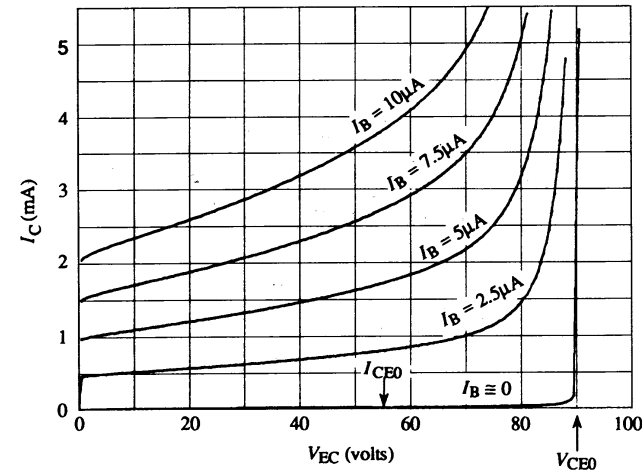
(a)



(b)



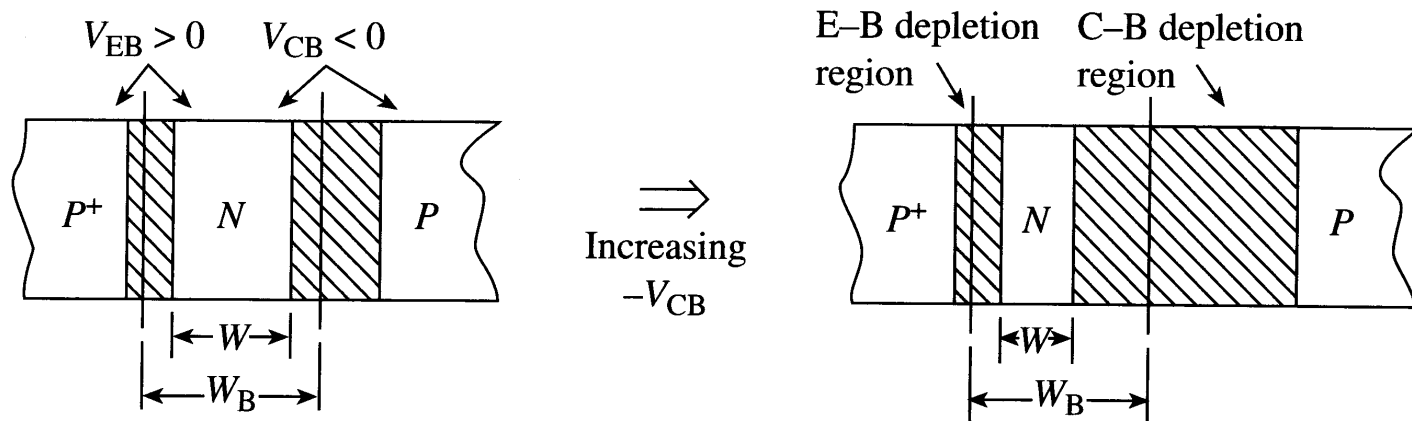
(c)



(d)

Base Width Modulation: “Early” Effect

- Base width has been assumed to be constant
- When bias voltages change, depletion widths change and the effective base width will be a function of the bias voltages
- Most of the effect comes from the C-B junction since the bias on the collector is usually larger than that on the E-B junction

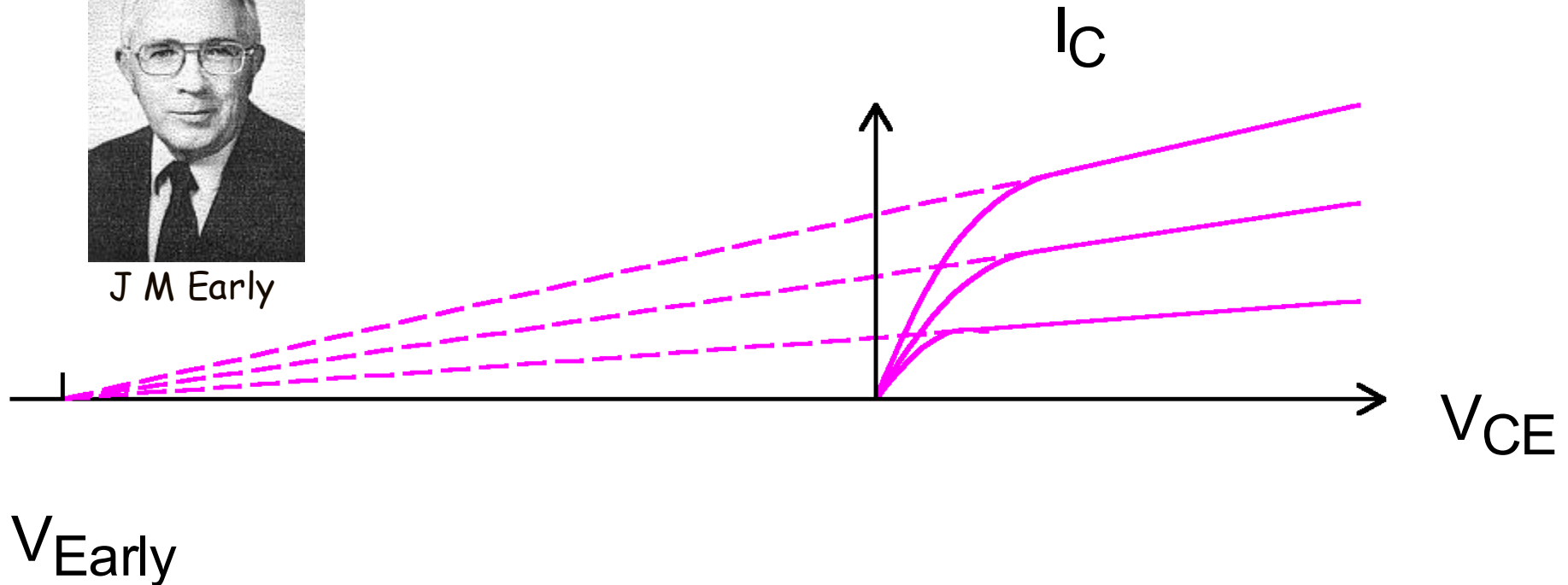


Base width gets smaller as applied voltages get larger

Early Voltage



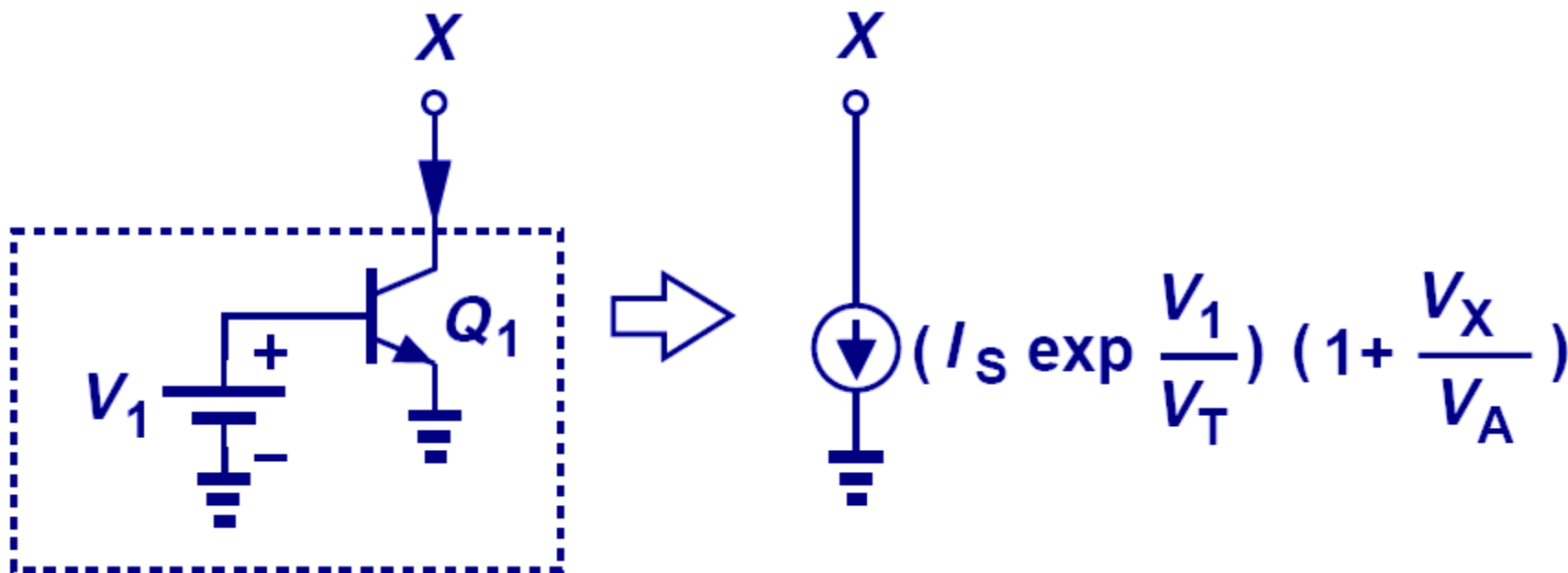
J M Early



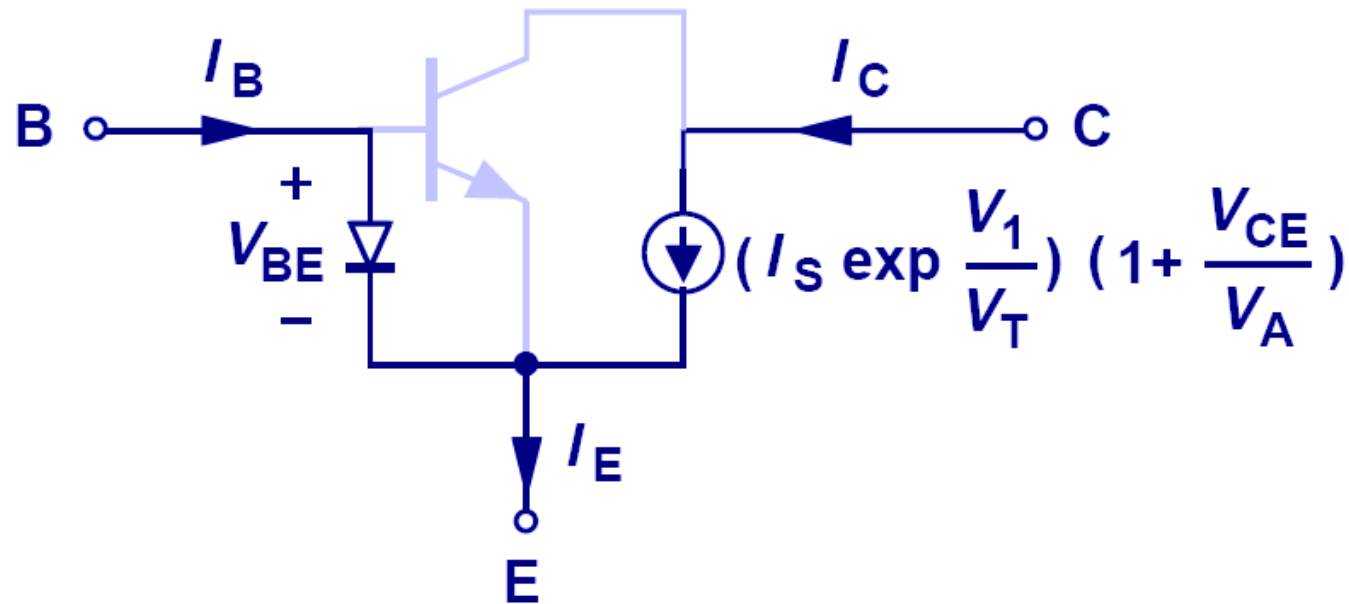
Converge ~ at single point called "Early Voltage" (after James Early)

Large "Early Voltage" = Absence of "Base Width Modulation"

Early Effect Representation

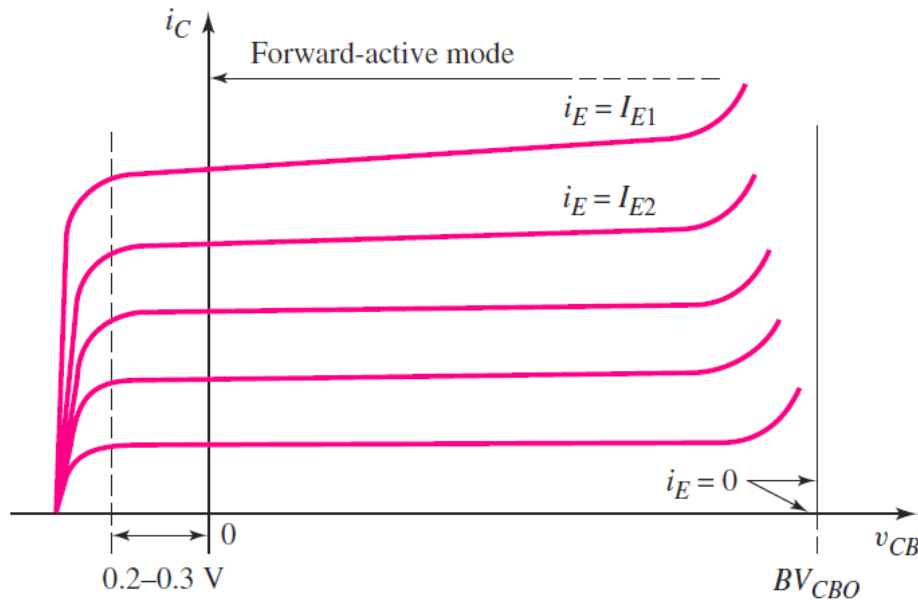


Early Effect and Large-Signal Model

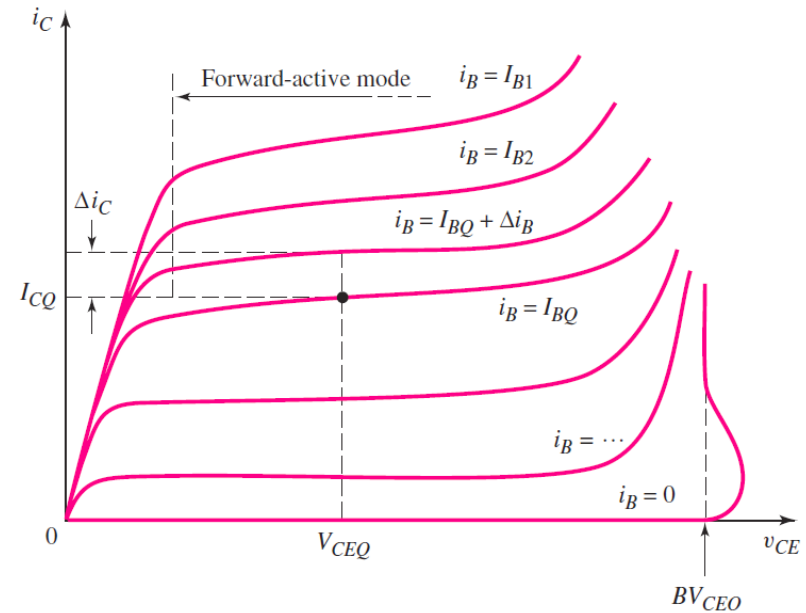


- Early effect can be accounted for in large-signal model by simply changing the collector current with a correction factor.
- In this model, base current does not change.

Other Second Order Effects: Breakdown Voltage



The i_C versus v_{CB} common-base characteristics, showing the collector–base junction breakdown



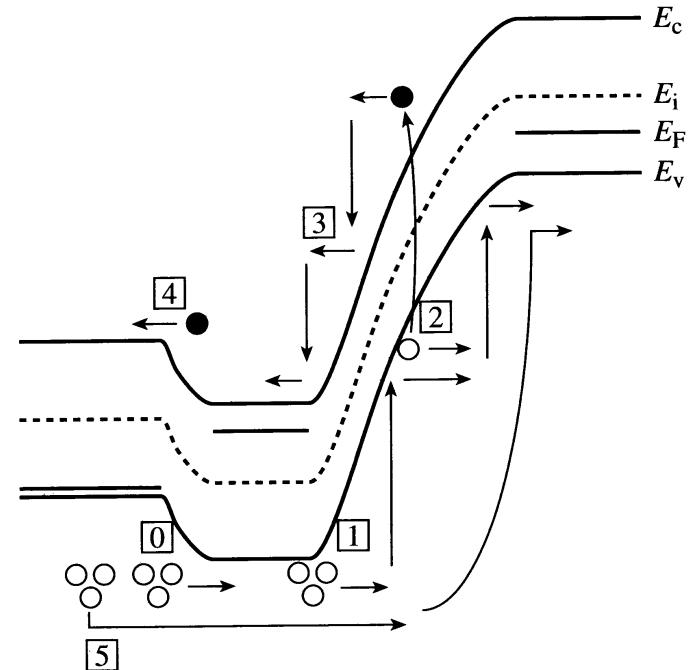
Common-emitter characteristics showing breakdown effects

$$BV_{CEO} = \frac{BV_{CBO}}{\sqrt[n]{\beta}}$$

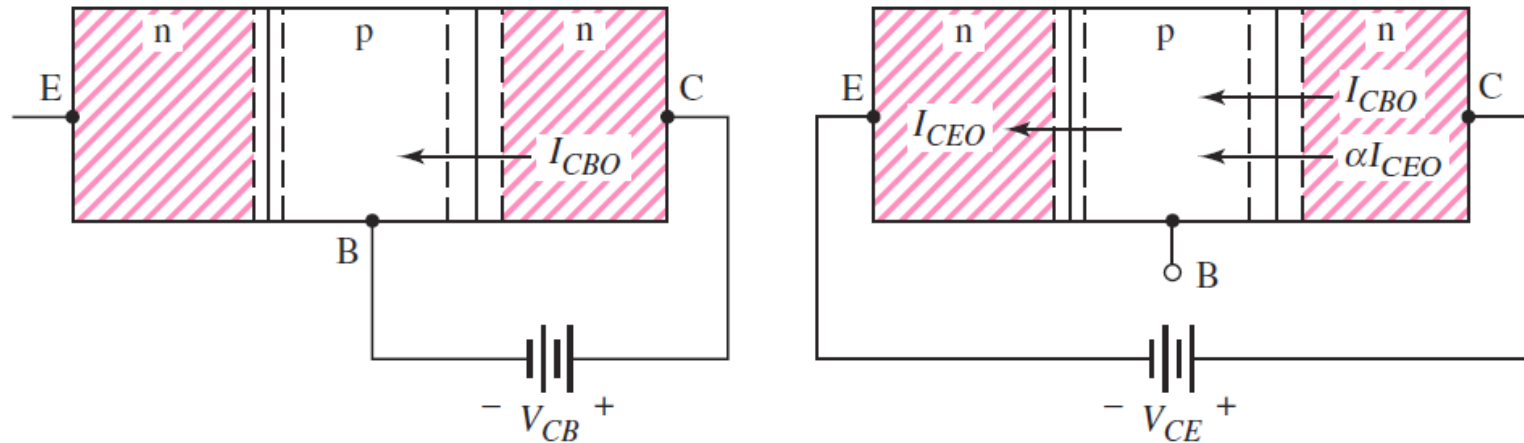
where n is an empirical constant usually in the range of 3 to 6.

Avalanche Multiplication Breakdown

- **Common Base:** Similar to single p-n junction $V_{CB0} \approx V_{BD}(B-C)$
- **Common Emitter:** more complicated
 1. holes injected by FB emitter to base
 2. holes generate e-p pairs in C-B depletion
 3. e^- drift back into base
 4. e^- injected to emitter
 5. more holes into base.....



Other Second Order Effects: Leakage Currents



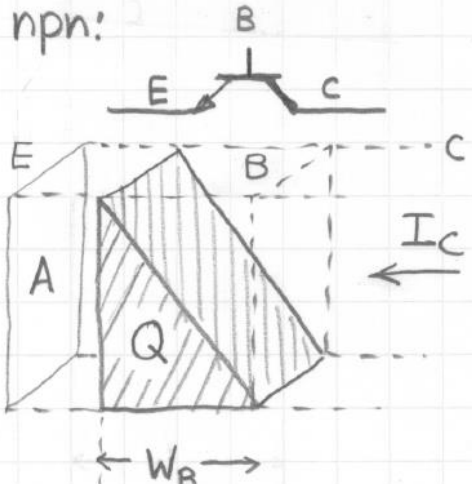
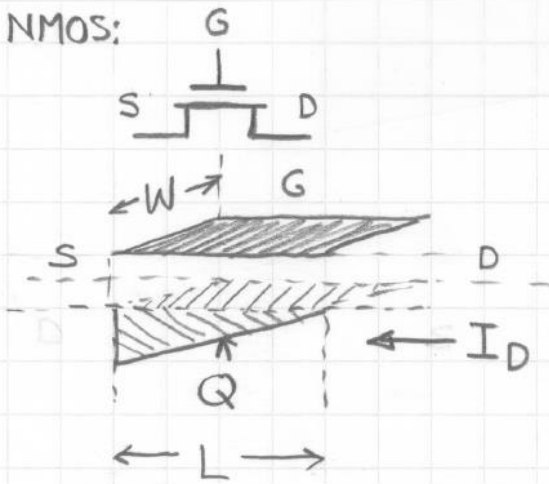
Block diagram of an npn transistor in an (a) open-emitter configuration showing the junction leakage current I_{CBO} and (b) open-base configuration showing the leakage current I_{CEO}

$$I_{CEO} = \alpha I_{CEO} + I_{CBO}$$

or

$$I_{CEO} = \frac{I_{CBO}}{1 - \alpha} \cong \beta I_{CBO}$$

BJT - MOS Comparison

npn: 	NMOS: 
Current in collector terminal: $I_C = \underbrace{\frac{qAD_n n_i^2}{2W_B N_A}}_{I_S} e^{(V_{BE}/V_T)}$	Current in drain terminal (active region): $I_D = \frac{\mu C_{ox}}{2} \frac{W}{L} (V_{GS} - V_t)^2$
Representing current in form of $I = Q/\Delta T$	
Charge in base: $Q = \frac{qAW_B n_i^2}{2N_A} e^{(V_{BE}/V_T)}$	Charge in channel: $Q = \frac{C_{ox}WL}{2} (V_{GS} - V_t)$
Transit time through base: $\Delta T = \frac{W_B^2}{2\mu V_T}$	Transit time through channel: $\Delta T = \frac{L^2}{2\mu(V_{GS} - V_t)}$

BJT - MOS Comparison

Speed

WB, L are critical dimensions for improving speed performance

Note exponent of 2 in ΔT equation indicates that improvement goes as factor squared; meaning

there are two reasons speed improves:

- 1) Shorter distance for carrier to travel
- 2) More "push" (steeper diffusion gradient for BJT, higher E field for MOSFET)

Factors in MOS - BJT speed performance:

- 1) Bulk mobility (BJT) always better than surface mobility (MOSFET)
- 2) Reducing critical dimension involves different process considerations
- 3) Trying to increase MOSFET speed by increasing $V_{GS} - V_t$ has two problems:
 - reduces transconductance
 - carrier velocity doesn't increase as much as expected due to velocity saturation

Small signal transconductance g_m :

$$g_m = \frac{I_C}{V_T}$$

$$g_m = \frac{I_D}{(V_{GS} - V_t)/2}$$

BJT

- More speed, transconductance per amount of bias current
- Lower noise for low R_S signal sources
- Higher intrinsic gain for actively loaded stage (better Early voltage)
- Lower output resistance at emitter vs. source of MOSFET (better buffer on output side)
- "Closer" to fundamental physics (e.g. bandgap voltage reference)
- Follows exponential model over 5 - 8 orders of magnitude (analog computation; multipliers)
- Higher output resistance current sources

- MOS

- ✓ Near ∞ input resistance looking into gate vs. base current for BJT (better buffer on input side)
- ✓ Lower noise for high R_S signal sources
- ✓ Better analog switch; truly ohmic at origin of V_{DS} -ID plot (sample & hold)
- ✓ Compatible with digital CMOS (process cost advantage)
- ✓ Comes out of non-active operating region more quickly (BJT slow out of saturation)
- ✓ More robust current sources (gentler "crash" than BJT into saturation)

BJT SPICE

BJT syntax

Q<name> <collector node> <base node> <emitter node> [<substrate node>] <modelname>

+ [<area>]

.MODEL <modelname> NPN(BF=<value> BR=<value> IS=<value> CJE=<value>

+ CJC=<value> VJE=<value> VJC=<value> VAF=<value> VAR=<value>

+ NF=<value> NR=<value>)

BJT Parameters

BF Forward active current gain

VJE Base-emitter built-in potential

BR Reverse active current gain

VJC Base-collector built-in potential

IS Transport saturation current

VAF Forward mode Early voltage

CJE Base-emitter zero-bias

VAR Reverse mode Early voltage

junction capacitance

NF Forward mode ideality factor

CJC Base-collector zero-bias

NR Reverse mode ideality factor

Junction capacitance

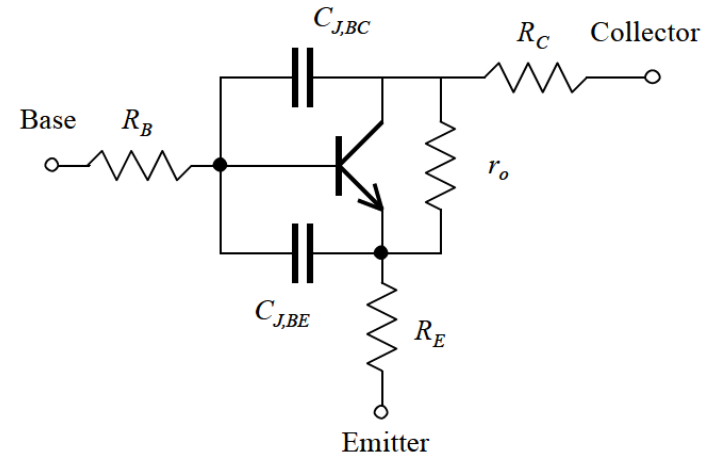


BJT SPICE Model

$$I_E = \frac{I_S}{\alpha_F} (e^{V_{BE}/V_T} - 1) - I_S (e^{V_{BC}/V_T} - 1)$$

$$I_C = I_S (e^{V_{BE}/V_T} - 1) - \frac{I_S}{\alpha_r} (e^{V_{BC}/V_T} - 1)$$

$$I_S = \alpha_F I_{ES} = \alpha_R I_{CS}$$



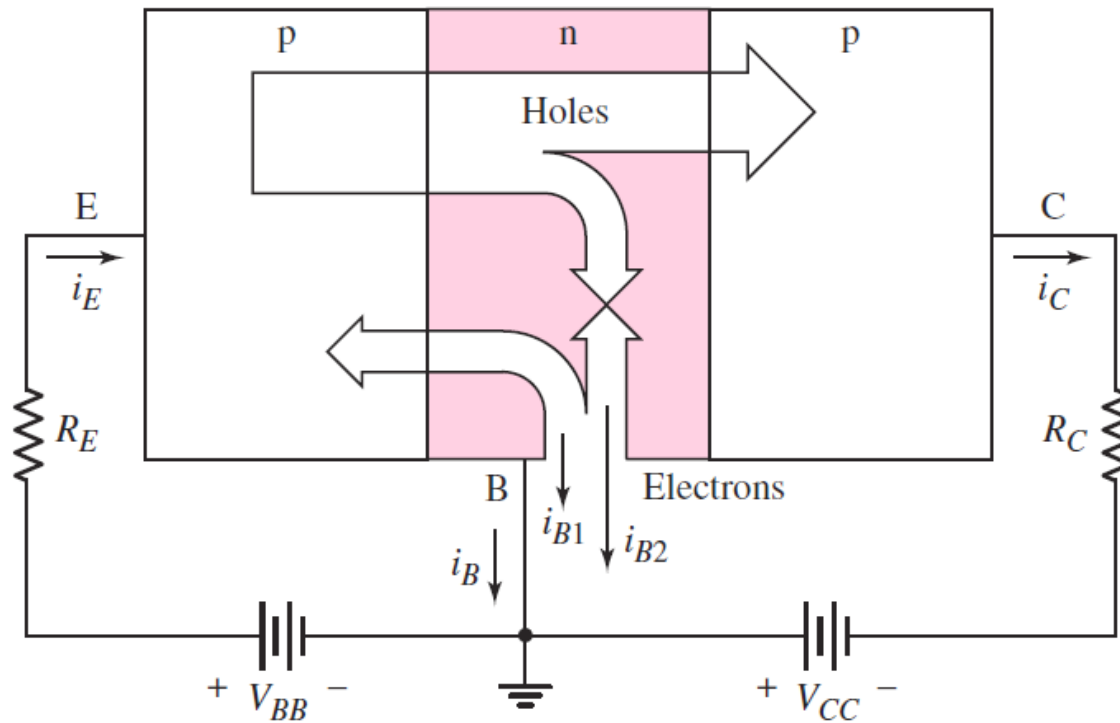
$$C_{BE} = \frac{dQ_{BE}}{dV_{BE}} = \frac{\tau_F I_S}{V_T} e^{V_{BE}/V_T} + \frac{C_{je0}}{[1 - (V_{BE}/\phi_e)]^{m_e}}$$

$$C_{BC} = \frac{dQ_{BC}}{dV_{BC}} = \frac{\tau_R I_S}{V_T} e^{V_{BC}/V_T} + \frac{C_{jc0}}{[1 - (V_{BC}/\phi_c)]^{m_c}}$$

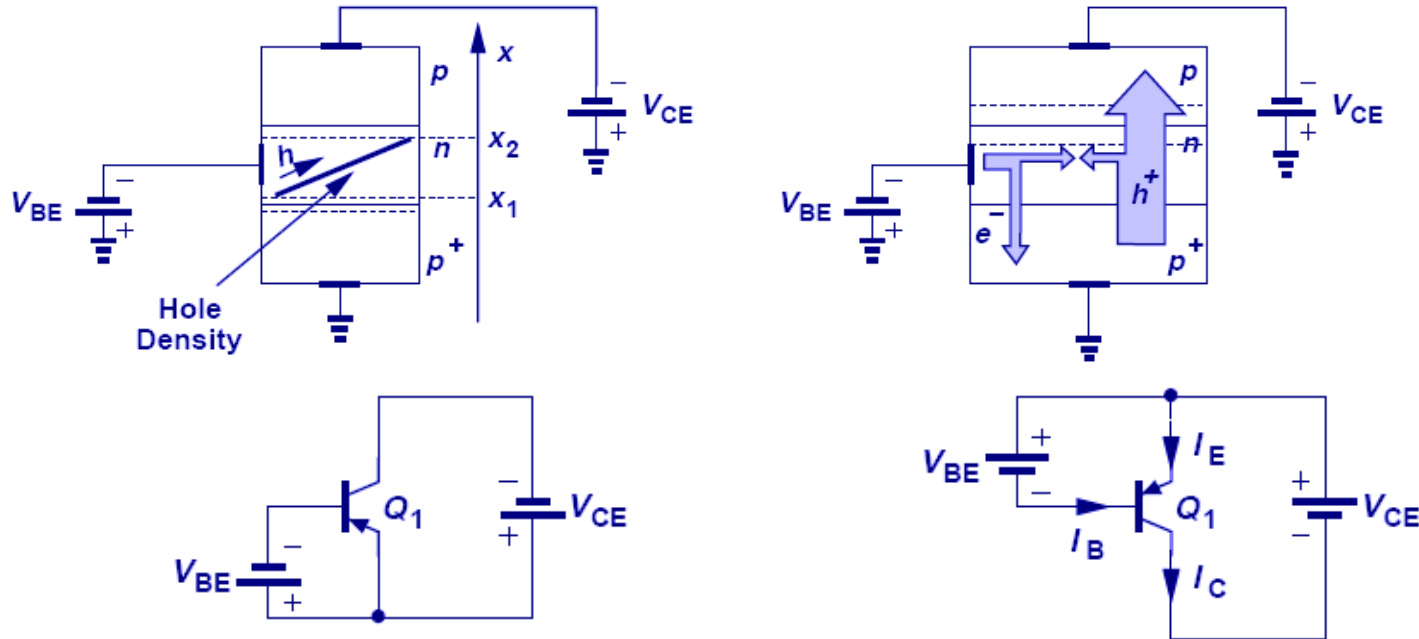
$$C_{CS} = \frac{C_{js0}}{[1 - (V_{SC}/\phi_s)]^{m_s}}$$

Go to Review Slides to review pn
junction capacitance calculations

Bipolar PNP Transistor

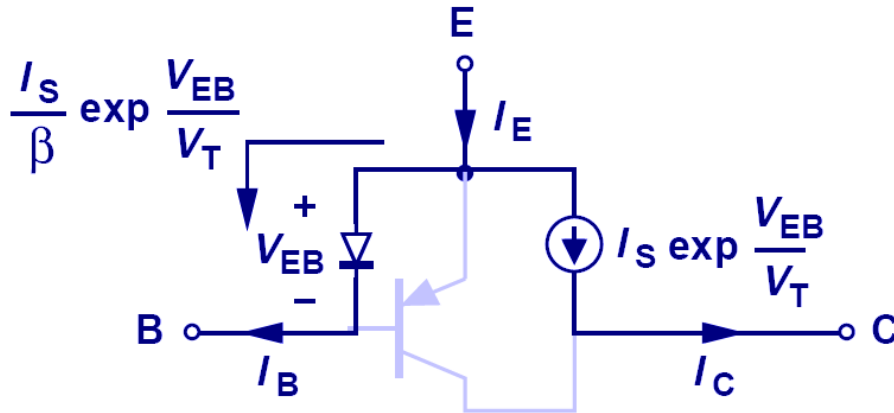


PNP Transistor



- With the polarities of emitter, collector, and base reversed, a PNP transistor is formed.
- All the principles that applied to NPN's also apply to PNP's, with the exception that emitter is at a higher potential than base and base at a higher potential than collector.

PNP Equations



$$I_C = I_S \exp \frac{V_{EB}}{V_T}$$

$$I_B = \frac{I_S}{\beta} \exp \frac{V_{EB}}{V_T}$$

$$I_E = \frac{\beta + 1}{\beta} I_S \exp \frac{V_{EB}}{V_T}$$

Early Effect



$$I_C = \left(I_S \exp \frac{V_{EB}}{V_T} \right) \left(1 + \frac{V_{EC}}{V_A} \right)$$

A Comparison between NPN and PNP

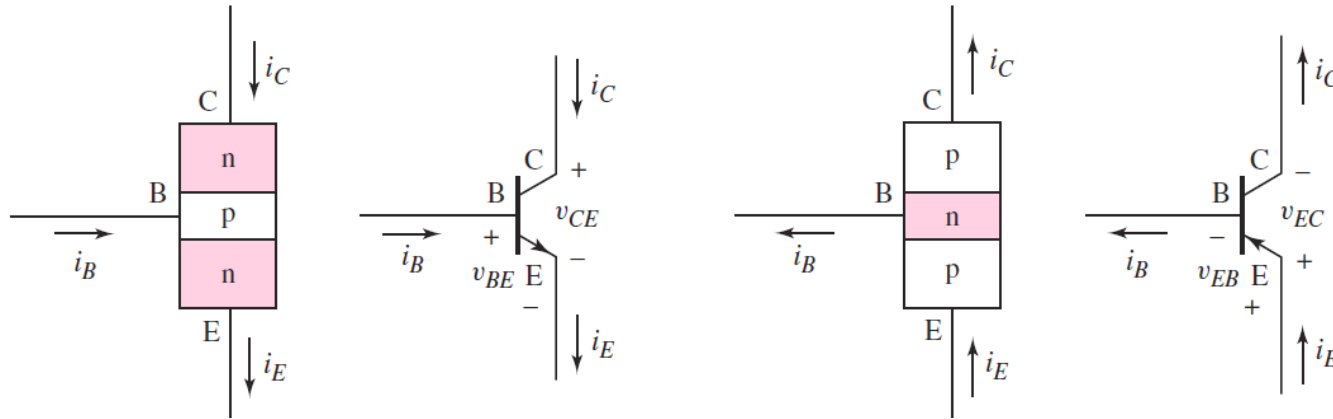
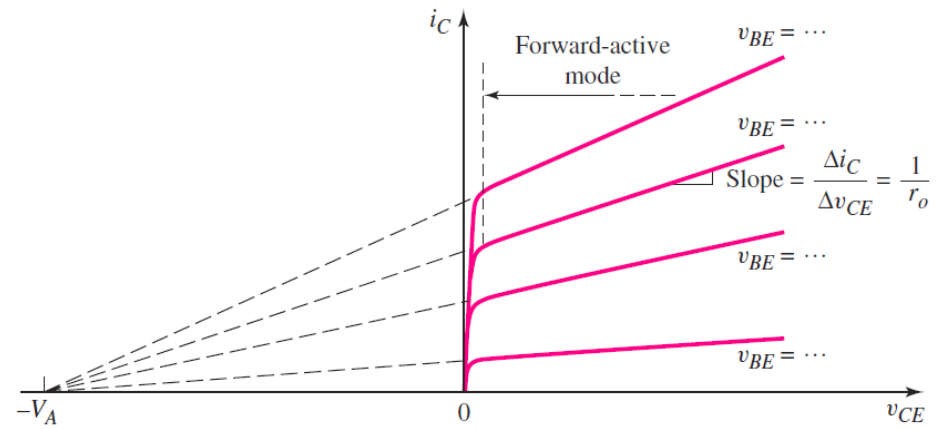
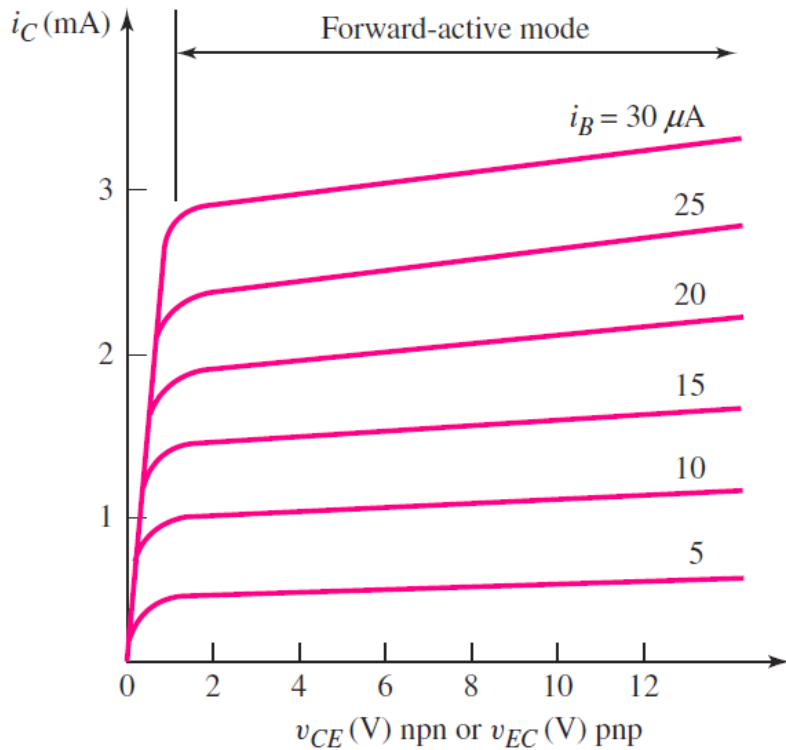


Table 5.1

Summary of the bipolar current-voltage relationships in the active region

npn	pnp
$i_C = I_S e^{v_{BE}/V_T}$	$i_C = I_S e^{v_{EB}/V_T}$
$i_E = \frac{i_C}{\alpha} = \frac{I_S}{\alpha} e^{v_{BE}/V_T}$	$i_E = \frac{i_C}{\alpha} = \frac{I_S}{\alpha} e^{v_{EB}/V_T}$
$i_B = \frac{i_C}{\beta} = \frac{I_S}{\beta} e^{v_{BE}/V_T}$	$i_B = \frac{i_C}{\beta} = \frac{I_S}{\beta} e^{v_{EB}/V_T}$
For both transistors	
$i_E = i_C + i_B$	$i_C = \beta i_B$
$i_E = (1 + \beta) i_B$	$i_C = \alpha i_E = \left(\frac{\beta}{1 + \beta} \right) i_E$
$\alpha = \frac{\beta}{1 + \beta}$	$\beta = \frac{\alpha}{1 - \alpha}$

I-V C/cs

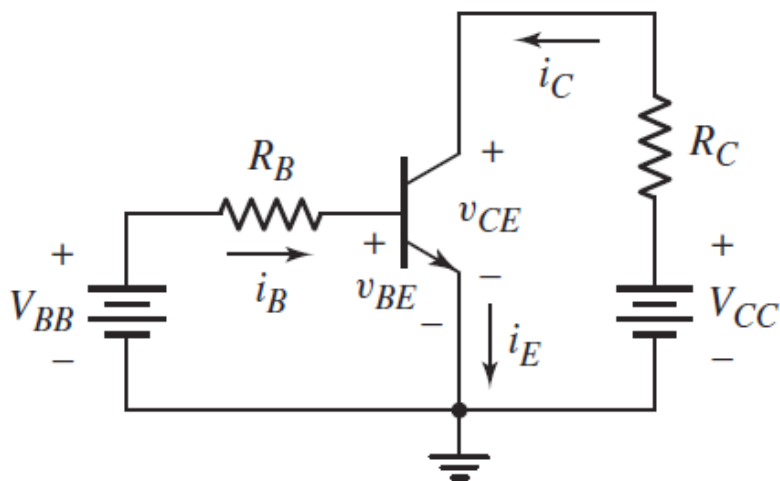


$$r_o \cong \frac{V_A}{I_C}$$

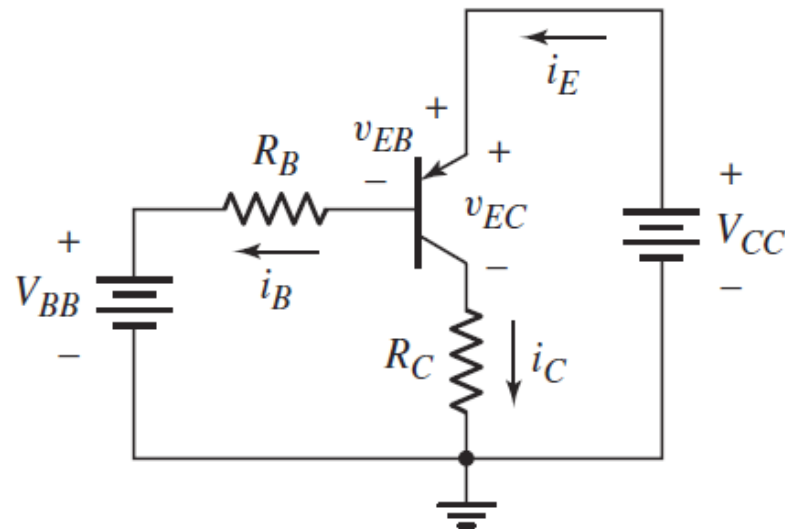
Test Your Understanding

TYU 5.4 The output resistance of a bipolar transistor is $r_o = 225 \text{ k}\Omega$ at $I_C = 0.8 \text{ mA}$.
(a) Determine the Early voltage. (b) Using the results of part (a), find r_o at
(i) $I_C = 0.08 \text{ mA}$ and (ii) $I_C = 8 \text{ mA}$.

NPN & PNP Biasing

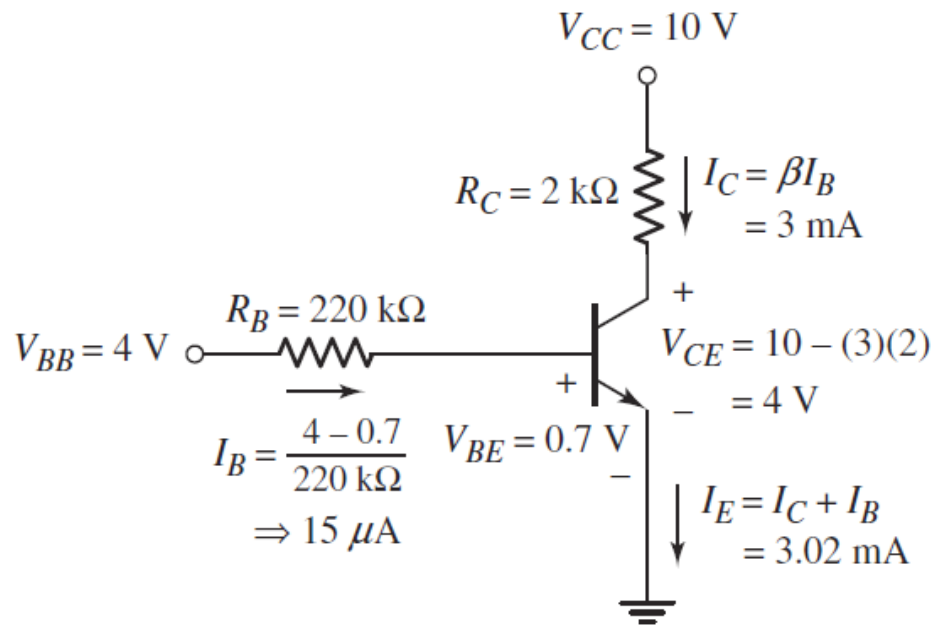
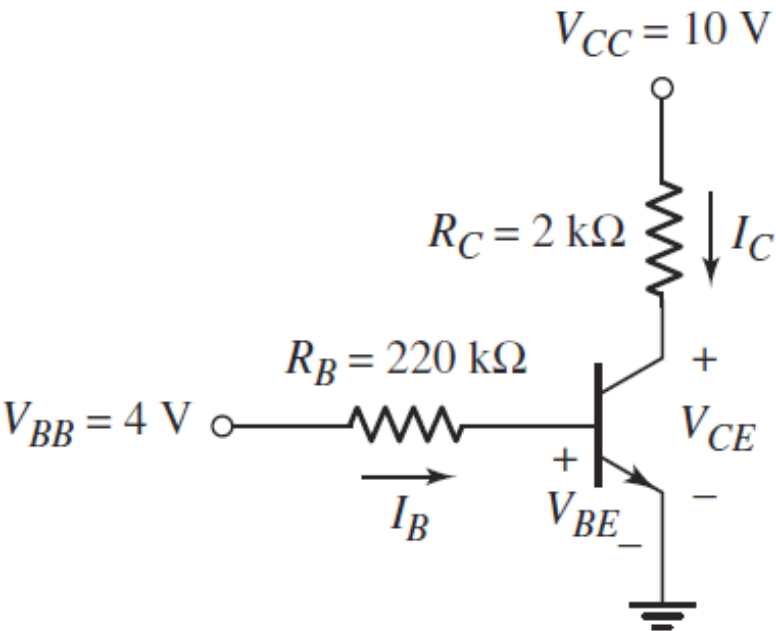


NPN: Collector is at a higher potential than both the base and Emitter.



PNP: Emitter is at a higher potential than both the base and collector.

DC-Analysis Examples

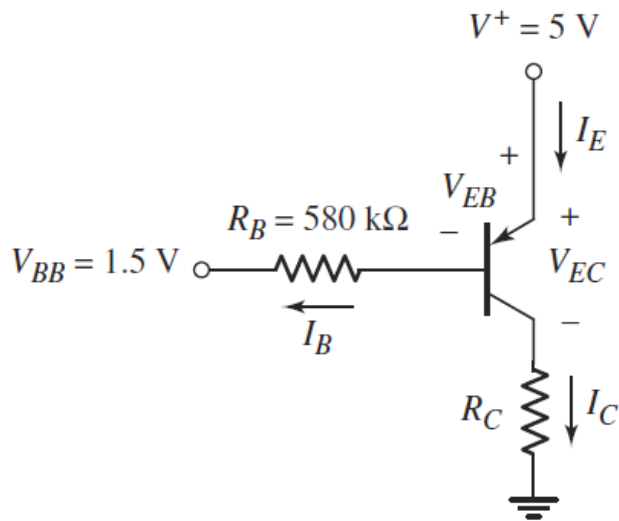


What is the power dissipated in the transistor?

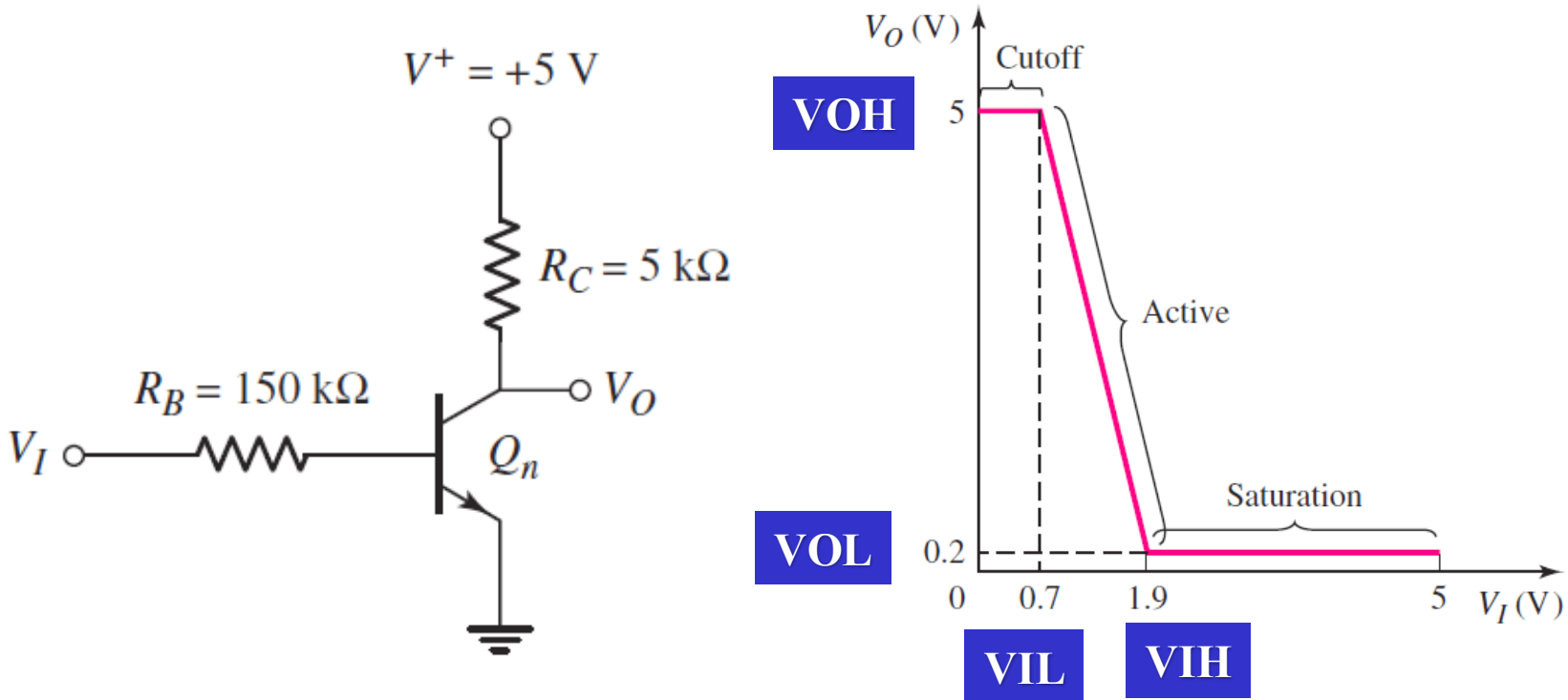
$$P_T = I_B V_{BE(\text{on})} + I_C V_{CE} = (0.015)(0.7) + (3)(4) \cong I_C V_{CE}$$

PNP DC Analysis Example

For the circuit shown in Figure 5.22(a), the parameters are: $V_{BB} = 1.5 \text{ V}$, $R_B = 580 \text{ k}\Omega$, $V^+ = 5 \text{ V}$, $V_{EB(\text{on})} = 0.6 \text{ V}$, and $\beta = 100$. Find I_B , I_C , I_E , and R_C such that $V_{EC} = (\frac{1}{2})V^+$. Use $V_{EB}=0.6\text{V}$

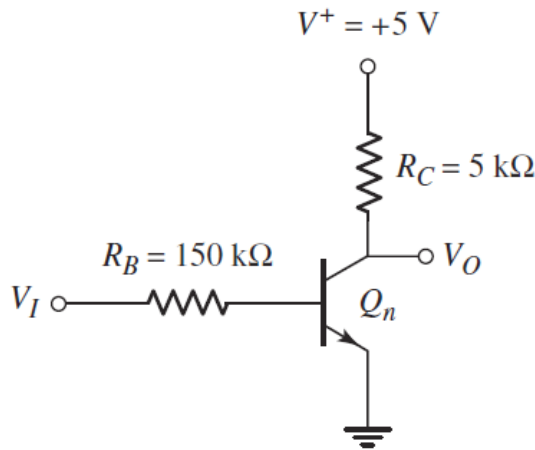


Voltage Transfer C/Cs (VTC)-NPN



Assume npn transistor parameters of $V_{BE}(\text{on}) = 0.7 \text{ V}$, $\beta = 120$, $V_{CE}(\text{sat}) = 0.2 \text{ V}$, and $V_A = \infty$,

Voltage Transfer C/Cs (VTC)-NPN



1. V_{OH} , which is equivalent to V_{CE} with the transistor at the edge of the cutoff region, that is, V_{CC} . For the given example it is 5 V. Hence,

$$V_{OH} = V_{CC}$$

However, V_{OH} is usually measured with $V_{in} = V_{OL}$

2. V_{OL} , which is equivalent to V_{CE} with the transistor at the edge of the saturation region, that is, $V_{CE(sat)}$. In the example it is 0.1 V. Thus,

$$V_{OL} = V_{CE(sat)}$$

V_{OL} is commonly measured with $V_{in} = V_{OH}$

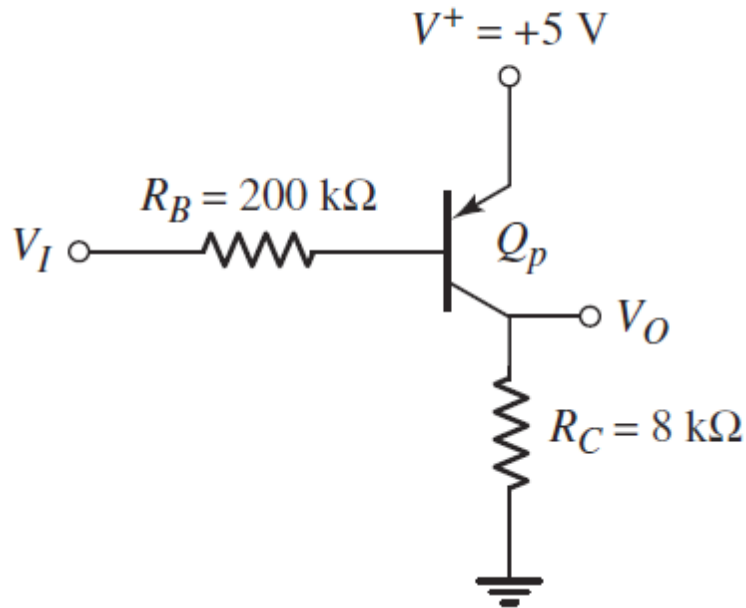
3. V_{IL} , which is the input voltage just sufficient to turn the transistor on. In this example it is $V_{BE(on)}$, which is given as 0.7 V. Therefore,

$$V_{IL} = V_{BE(on)}$$

4. V_{IH} , which is the input voltage just sufficient to saturate the transistor. Now with the transistor just at the edge of saturation, $I_C = I_{C(EOS)}$, where

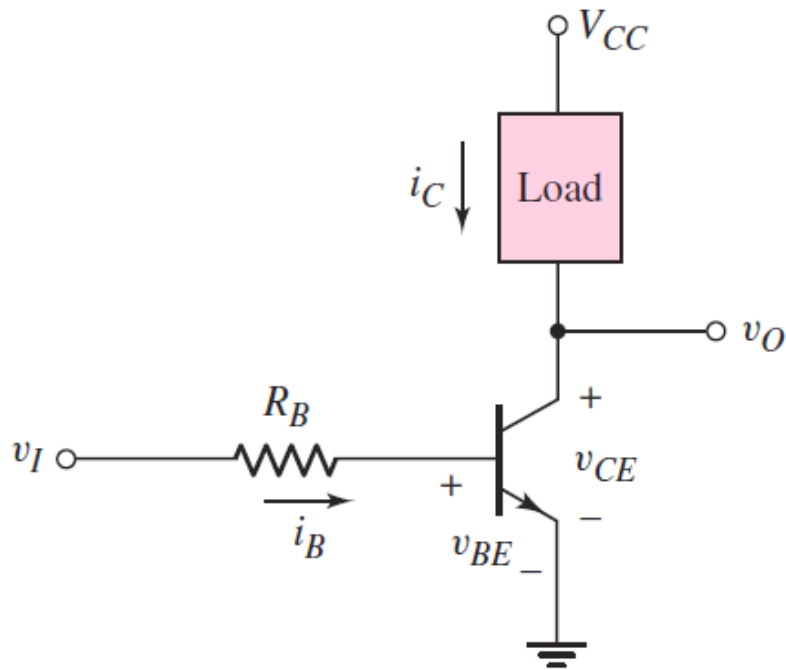
$$V_{IH} = V_{BE(sat)} + \frac{R_B}{R_C} \frac{V_{CC} - V_{CE(sat)}}{\beta_F}$$

Voltage Transfer C/Cs (VTC)-NPN



Assume npn transistor parameters of $V_{EB}(\text{on}) = 0.7 \text{ V}$, $\beta = 80$, $V_{EC}(\text{sat}) = 0.2 \text{ V}$, and $V_A = \infty$.

BJT As a Switch



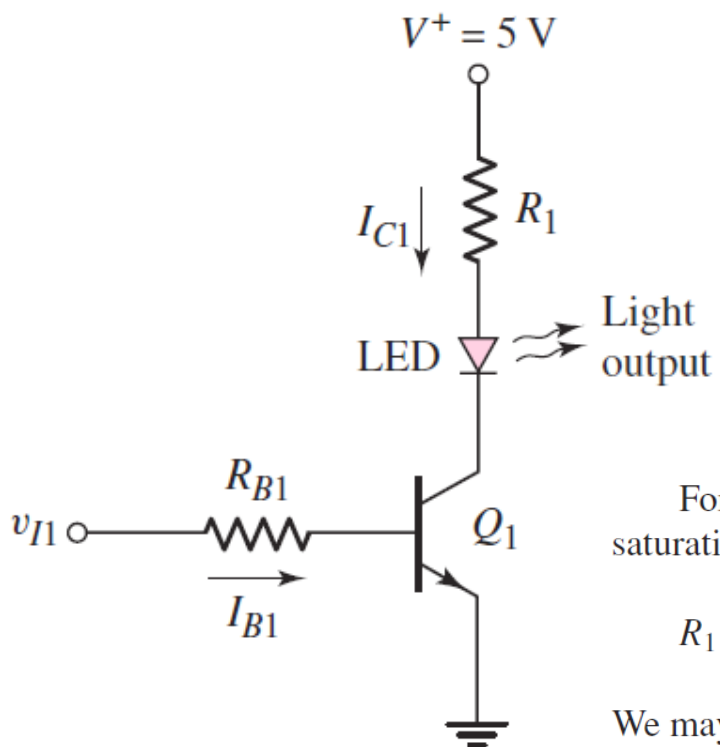
$$i_B \cong \frac{v_I - V_{BE}(\text{on})}{R_B}$$

$$i_C = I_C(\text{sat}) = \frac{V_{CC} - V_{CE}(\text{sat})}{R_C}$$

and

$$v_O = V_{CE}(\text{sat})$$

Example



The transistor in the inverter is used to turn the light-emitting diode (LED) on and off. The required LED current is $I_{C1} = 12$ mA to produce the specified output light. Assume transistor parameters of $\beta = 80$, $V_{BE}(\text{on}) = 0.7$ V, and $V_{CE}(\text{sat}) = 0.2$ V, and assume the diode cut-in voltage is $V_\gamma = 1.5$ V. [Note: LEDs are fabricated with compound semiconductor materials and have a larger cut-in voltage compared to silicon diodes.]

For $v_{I1} = 5$ V, we require $I_{C1} = 12$ mA and want the transistor to be driven into saturation. Then

$$R_1 = \frac{V^+ - (V_\gamma + V_{CE}(\text{sat}))}{I_{C1}} = \frac{5 - (1.5 + 0.2)}{12} \Rightarrow R_1 = 275 \Omega$$

We may let $I_{C1}/I_{B1} = 40$. Then $I_{B1} = 12/40 = 0.3$ mA. Now

$$R_{B1} = \frac{v_{I1} - V_{BE}(\text{on})}{I_{B1}} = \frac{5 - 0.7}{0.3} = 14.3 \text{ k}\Omega$$

The power dissipation in Q_1 is

$$P_1 = I_{B1}V_{BE}(\text{on}) + I_{C1}V_{CE}(\text{sat}) = (0.3)(0.7) + (12)(0.2) = 2.61 \text{ mW}$$

Digital Gates

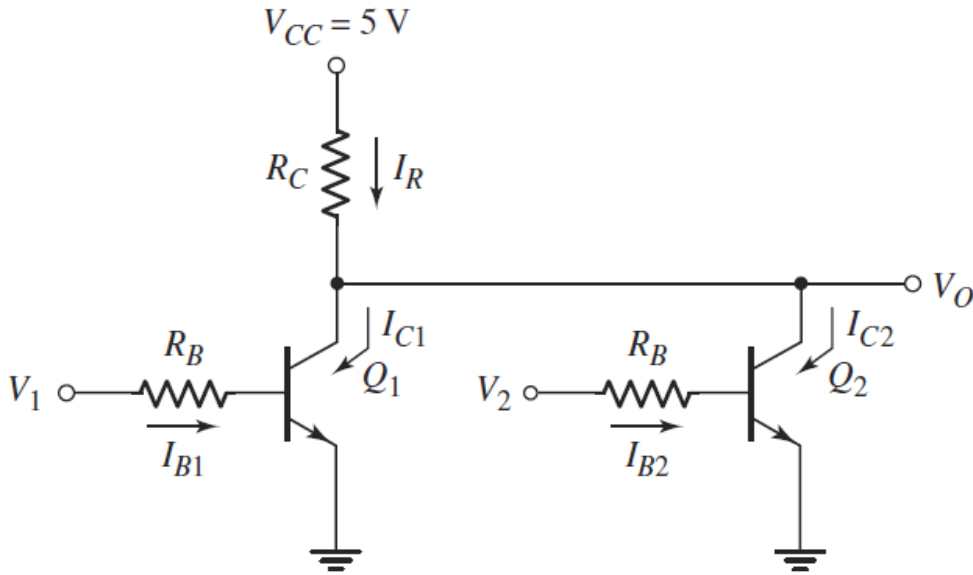


Table 5.2 The bipolar NOR logic circuit response

V_1 (V)	V_2 (V)	V_O (V)
0	0	5
5	0	0.2
0	5	0.2
5	5	0.2

Condition	V_O	I_R	Q_1	Q_2
$V_1 = 0$, $V_2 = 0$	5 V	0	$I_{B1} = I_{C1} = 0$	$I_{B2} = I_{C2} = 0$
$V_1 = 5$ V, $V_2 = 0$	0.2 V	$\frac{5 - 0.2}{1} = 4.8$ mA	$I_{B1} = \frac{5 - 0.7}{20} = 0.215$ mA $I_{C1} = I_R = 4.8$ mA	$I_{B2} = I_{C2} = 0$
$V_1 = 0$, $V_2 = 5$ V	0.2 V	4.8 mA	$I_{B1} = I_{C1} = 0$	$I_{B2} = 0.215$ mA $I_{C2} = I_R = 4.8$ mA
$V_1 = 5$ V, $V_2 = 5$ V	0.2 V	4.8 mA	$I_{B1} = 0.215$ mA $I_{C1} = \frac{I_R}{2} = 2.4$ mA	$I_{B2} = 0.215$ mA $I_{C2} = \frac{I_R}{2} = 2.4$ mA